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SKETCH-AND-RESTART: RANDOMIZED SKETCHING IN QUADRATURE-BASED RESTARTING FOR MATRIX FUNCTIONS

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Abstract. We develop a sketch-and-restart framework for computing the action of a matrix function on a vector, $f(\boldsymbol{A})\boldsymbol{b}$, where $\boldsymbol{A}$ is large, sparse, and non-Hermitian. The framework combines quadrature-based restarting with Arnoldi-like decompositions generated by sketched or truncated Arnoldi processes. Within this framework, we develop two classes of restarted algorithms. The first uses a fixed Krylov subspace dimension and is based either on the sketched Arnoldi process or on a new sketched harmonic Arnoldi process proposed in this work. The second class chooses the Krylov subspace dimension adaptively by running the truncated Arnoldi process until the condition number of the generated basis, estimated from its sketch, exceeds a prescribed threshold. We also establish the convergence of the restarted sketched harmonic Arnoldi method for Stieltjes functions under the assumption that $\boldsymbol{A}$ is positive real. Numerical experiments demonstrate the effectiveness of the proposed framework, including the computational savings achieved through sketching, the storage reduction enabled by adaptive truncation, and the acceleration obtained from thick restarting.

Key words. matrix function, Krylov method, sketching, restarting

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1. Introduction.

This paper considers the computation of $f(\boldsymbol{A})\boldsymbol{b}$, where $\boldsymbol{A} \in \mathbb{C}^{N\times N}$ is a matrix, $\boldsymbol{b} \in \mathbb{C}^N$ is a vector, and f is a suitable scalar function. Such computations arise in many areas of scientific computing.

If $\boldsymbol{A}$ is small and dense, one can first compute $f(\boldsymbol{A})$ explicitly using algorithms such as those described in [16], and then apply it to $\boldsymbol{b}$. For large matrices, however, this approach is infeasible because of its computational and memory costs. *Krylov subspace methods* are therefore commonly used to approximate $f(\boldsymbol{A})\boldsymbol{b}$, as they require only matrix-vector products with $\boldsymbol{A}$, which can often be computed efficiently when $\boldsymbol{A}$ is sparse. Their main limitation is the need to store and orthogonalize the Krylov basis. For non-Hermitian matrices, the Arnoldi process requires $\mathscr{O}(Nm^2)$ operations to generate m orthonormal basis vectors and $\mathscr{O}(Nm)$ storage for the basis, and these costs can become substantial when m is large.

Many approaches have been proposed to address these limitations. One important class is based on *restarting*, in which only a fixed number of basis vectors need to be stored [1, 9]. The *quadrature-based restarting* method of [12] uses an integral representation of f, together with interpolation polynomials, to derive an integral representation of the error. Approximating this error representation by quadrature allows the process to be restarted without storing all previously generated basis vectors. However, restarted methods may converge slowly, and this issue can be alleviated by *thick restarting* strategies [4, 10, 18].

Another class of approaches is based on *randomized sketching* [6,15]. These methods approximate $f(\boldsymbol{A})\boldsymbol{b}$ using a nonorthonormal basis of a Krylov subspace, often generated by the sketched Arnoldi process [2] or the truncated Arnoldi process [19]. The sketched Arnoldi process computes the inner products required for orthogonaliza-

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tion in the sketched space, thereby reducing the orthogonalization cost by nearly one half. The truncated Arnoldi process reduces the orthogonalization cost to $\mathcal{O}(Nmt)$, where t is the truncation parameter. However, the condition number of the generated basis may grow exponentially [3], which limits the usable Krylov subspace dimension.

We focus on the case in which $\boldsymbol{A}$ is large, sparse, and non-Hermitian. We develop a sketch-and-restart framework for computing $f(\boldsymbol{A})\boldsymbol{b}$ by combining randomized sketching with quadrature-based restarting. The Krylov methods considered here, including their sketched variants, approximate $f(\boldsymbol{A})\boldsymbol{b}$ by $p_{m-1}(\boldsymbol{A})\boldsymbol{b}$ in $\operatorname{span}\{\boldsymbol{b}, \boldsymbol{A}\boldsymbol{b}, \ldots, \boldsymbol{A}^{m-1}\boldsymbol{b}\}$. The polynomial p_{m-1} has degree at most $m-1$ and interpolates f at the eigenvalues of the $m \times m$ reduced matrix obtained by an Arnoldi-type process. This representation allows us to apply the existing integral error representation and quadrature-based restarting of [12], with the corresponding projection conditions specified in Section 2.

Within this framework, we develop two classes of restarted algorithms. The first uses a fixed Krylov subspace dimension and is based either on the existing sketched Arnoldi process or on a new sketched harmonic Arnoldi process proposed in this work. The sketched harmonic Arnoldi process generates a nonorthonormal Krylov basis satisfying the sketched harmonic orthogonality condition needed for the restarting framework. These processes lead, respectively, to the restarted sketched Arnoldi (sArn) and restarted sketched harmonic Arnoldi (sHmArn) methods. The second class comprises adaptive restarted Krylov methods. The Krylov subspace dimension is selected by running the truncated Arnoldi process until the condition number of the generated basis, estimated from its sketch, exceeds a prescribed threshold. A rank-1 update enforces the Euclidean Arnoldi or harmonic Arnoldi orthogonality condition, yielding adaptive restarted Arnoldi (adaArn-t) or adaptive restarted harmonic Arnoldi (adaHmArn-t), respectively. For Stieltjes functions, we relate the non-sketched adaptive variants to the existing convergence results of [11] and derive conditional convergence bounds for sHmArn when $\boldsymbol{A}$ is positive real.

Recently, Guidotti et al. [14] combine randomized Arnoldi with restarting by evaluating the function of a block Hessenberg matrix that grows with the number of restart cycles. We instead evaluate restart corrections by quadrature using an integral error representation on the current projected matrix. For identical cycle decompositions and exact quadrature, their block formulation and our fixed-dimension restarted sArnoldi method yield the same approximants in exact arithmetic.

The remainder of this paper is organized as follows. Section 2 formulates Krylov and sketched Krylov approximations within a common Arnoldi-like framework. Section 3 extends quadrature-based restarting to this framework. Section 4 presents the proposed algorithms, while Section 5 analyzes their convergence. Numerical experiments are presented in Section 6. Finally, Section 7 concludes the paper.

We use the following notation. The inner product of two vectors $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^N$ is defined as $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \boldsymbol{y}^* \boldsymbol{x}$. The open right half-plane is denoted by $\mathbb{C}_+ = \{z \in \mathbb{C} : \Re(z) > 0\}$. The symbol $\|\cdot\|$ denotes the Euclidean norm for vectors and the spectral norm for matrices. For a matrix $\boldsymbol{A}$, we denote its spectrum by $\Lambda(\boldsymbol{A})$, its condition number by $\kappa(\boldsymbol{A}) = \|\boldsymbol{A}\|\|\boldsymbol{A}^{-1}\|$, and its numerical range by

$$\mathcal{W}(\boldsymbol{A}) := \{\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x} : \boldsymbol{x} \in \mathbb{C}^N,\ \|\boldsymbol{x}\| = 1\}.$$

A matrix $\boldsymbol{A} \in \mathbb{C}^{N \times N}$ is called positive real if $\Re(\boldsymbol{x}^* \boldsymbol{A} \boldsymbol{x}) > 0$ for every nonzero $\boldsymbol{x} \in \mathbb{C}^N$. A matrix $\boldsymbol{V} \in \mathbb{C}^{N \times m}$ is called orthonormal if its columns form an orthonormal set. For a vector $\boldsymbol{x}$ and a subspace $\mathcal{V}$, we write $\boldsymbol{x} \perp \mathcal{V}$ if $\boldsymbol{x}$ is orthogonal to $\mathcal{V}$. If $\boldsymbol{V}$ is a basis of $\mathcal{V}$, this condition is equivalent to $\boldsymbol{V}^* \boldsymbol{x} = \boldsymbol{0}$, and we also write $\boldsymbol{x} \perp \boldsymbol{V}$. The

set of polynomials of degree at most m is denoted by $\mathcal{P}_m$.

2. Krylov subspace methods and their sketched variants. This section places several Krylov and sketched Krylov approximations within a common Arnoldi-like framework. Specifically, we show that the Arnoldi and harmonic Arnoldi approximations, together with their sketched variants, can all be expressed in the form $\boldsymbol{B}_m f(\boldsymbol{H}_m)\boldsymbol{e}_1\beta$, where $\boldsymbol{B}_m$ is a possibly nonorthonormal basis of a Krylov subspace and $\boldsymbol{H}_m$ is an upper Hessenberg matrix arising from an Arnoldi-like decomposition. This observation forms the algebraic foundation for the quadrature-based restarting mechanism developed in Section 3.

Suppose that f is analytic on the closure of a domain $\Omega \subset \mathbb{C}$ containing $\Lambda(\boldsymbol{A})$, and admits the integral representation

$$f(z) = \int_\Gamma \frac{g(t)}{t-z}\,\mathrm{d}t, \qquad z \in \Omega, \tag{2.1}$$

where $\Gamma \subset \mathbb{C} \setminus \Omega$ is a contour and g is a suitable function on Γ. Such a representation follows, for example, from the Cauchy integral formula, in which case Γ encloses $\Lambda(\boldsymbol{A})$, and also includes Stieltjes functions such as $f(z) = z^{-\alpha}$ with $0 < \alpha < 1$; see also [12]. Then

$$f(\boldsymbol{A})\boldsymbol{b} = \int_\Gamma g(t)(t\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{b}\,\mathrm{d}t = \int_\Gamma g(t)\boldsymbol{x}(t)\,\mathrm{d}t, \tag{2.2}$$

where $\boldsymbol{x}(t)$ solves the shifted linear system

$$(t\boldsymbol{I} - \boldsymbol{A})\boldsymbol{x}(t) = \boldsymbol{b}. \tag{2.3}$$

Let $\mathcal{K}_{m+1}(\boldsymbol{A},\boldsymbol{b}) = \operatorname{span}\{\boldsymbol{b}, \boldsymbol{A}\boldsymbol{b}, \ldots, \boldsymbol{A}^m\boldsymbol{b}\}$ be the Krylov subspace generated by $\boldsymbol{A}$ and $\boldsymbol{b}$. Throughout this paper, we assume that the Krylov subspaces are nondegenerate, i.e., $\dim \mathcal{K}_j(\boldsymbol{A},\boldsymbol{b}) = j$ for $1 \le j \le m+1$. An associated *Arnoldi-like decomposition* is

$$\begin{aligned} \boldsymbol{A}\boldsymbol{B}_m &= \boldsymbol{B}_m\boldsymbol{H}_m + \boldsymbol{b}_{m+1}h_{m+1,m}\boldsymbol{e}_m^\top \\ &= [\boldsymbol{B}_m\ \boldsymbol{b}_{m+1}] \begin{bmatrix} \boldsymbol{H}_m \\ h_{m+1,m}\boldsymbol{e}_m^\top \end{bmatrix} =: \boldsymbol{B}_{m+1}\underline{\boldsymbol{H}}_m, \end{aligned} \tag{2.4}$$

where, for $1 \le j \le m+1$, $\boldsymbol{B}_j = [\boldsymbol{b}_1\ \cdots\ \boldsymbol{b}_j] \in \mathbb{C}^{N\times j}$ is a basis of $\mathcal{K}_j(\boldsymbol{A},\boldsymbol{b})$. Consequently, the matrix $\boldsymbol{H}_m \in \mathbb{C}^{m\times m}$ is upper Hessenberg. We assume that $\boldsymbol{b} = \boldsymbol{b}_1\beta$ for some $\beta \in \mathbb{C}$. When $\boldsymbol{B}_{m+1}$ is orthonormal, the Arnoldi-like decomposition (2.4) reduces to the standard *Arnoldi decomposition*, which can be computed by the Arnoldi process [21]. Throughout this paper, the data $\boldsymbol{B}_m$, $\boldsymbol{H}_m$, $\boldsymbol{b}_{m+1}$, $h_{m+1,m}$, and β in (2.4) depend on the chosen method.

2.1. Krylov subspace methods. In Krylov subspace methods, the solution $\boldsymbol{x}(t)$ of the shifted linear system (2.3) is approximated in $\mathcal{K}_m(\boldsymbol{A},\boldsymbol{b})$ by $\boldsymbol{x}_m(t) = \boldsymbol{B}_m\boldsymbol{y}_m(t)$, where $\boldsymbol{y}_m(t) \in \mathbb{C}^m$. The corresponding residual is $\boldsymbol{r}_m(t) = \boldsymbol{b} - (t\boldsymbol{I} - \boldsymbol{A})\boldsymbol{x}_m(t)$. The coefficient vector $\boldsymbol{y}_m(t)$ is determined by imposing a *Petrov–Galerkin condition* on $\boldsymbol{r}_m(t)$.

If the projection condition is $\boldsymbol{r}_m(t) \perp \mathcal{K}_m(\boldsymbol{A},\boldsymbol{b})$, then $\boldsymbol{x}_m^{\mathrm{Arn}}(t)$ is the Arnoldi approximation to the solution of the shifted linear system (2.3). This is equivalent to solving the shifted linear system using the full orthogonalization method (FOM) [21]. Solving the resulting reduced system gives

$$\boldsymbol{y}_m^{\mathrm{Arn}}(t) = \left(t\boldsymbol{I}_m - \boldsymbol{B}_m^\dagger\boldsymbol{A}\boldsymbol{B}_m\right)^{-1}\boldsymbol{e}_1\beta.$$

Replacing $\boldsymbol{x}(t)$ in (2.2) by $\boldsymbol{x}_m^{\mathrm{Arn}}(t)$, we obtain

$$\begin{aligned}\boldsymbol{f}_m^{\mathrm{Arn}} &= \int_\Gamma g(t)\boldsymbol{x}_m^{\mathrm{Arn}}(t)\,\mathrm{d}t = \boldsymbol{B}_m\Big(\int_\Gamma g(t)\big(t\boldsymbol{I}_m - \boldsymbol{B}_m^\dagger\boldsymbol{A}\boldsymbol{B}_m\big)^{-1}\,\mathrm{d}t\Big)\boldsymbol{e}_1\beta \\ &= \boldsymbol{B}_m f\big(\boldsymbol{B}_m^\dagger\boldsymbol{A}\boldsymbol{B}_m\big)\boldsymbol{e}_1\beta,\end{aligned} \tag{2.5}$$

provided that the corresponding matrix function is well-defined. The vector $\boldsymbol{f}_m^{\mathrm{Arn}}$ in (2.5) is called the *Arnoldi approximation* to $f(\boldsymbol{A})\boldsymbol{b}$. If $\boldsymbol{b}_{m+1} \perp \boldsymbol{B}_m$ in the Arnoldi-like decomposition (2.4), then

$$\boldsymbol{B}_m^\dagger\boldsymbol{A}\boldsymbol{B}_m = \boldsymbol{H}_m \quad \text{and} \quad \boldsymbol{f}_m^{\mathrm{Arn}} = \boldsymbol{B}_m f(\boldsymbol{H}_m)\boldsymbol{e}_1\beta. \tag{2.6}$$

Thus, under this orthogonality condition, the Arnoldi approximation can be computed directly from the upper Hessenberg matrix $\boldsymbol{H}_m$. This condition holds, in particular, when $\boldsymbol{B}_{m+1}$ is generated by the standard Arnoldi process.

If the projection condition is $\boldsymbol{r}_m(t) \perp \boldsymbol{A}\mathcal{K}_m(\boldsymbol{A},\boldsymbol{b})$, then $\boldsymbol{x}_m^{\mathrm{HmArn}}(t)$ is the harmonic Arnoldi approximation to the solution of the shifted linear system (2.3). By a similar argument,

$$\boldsymbol{y}_m^{\mathrm{HmArn}}(t) = \Big(t\boldsymbol{I}_m - \big((\boldsymbol{A}\boldsymbol{B}_m)^*\boldsymbol{B}_m\big)^{-1}(\boldsymbol{A}\boldsymbol{B}_m)^*(\boldsymbol{A}\boldsymbol{B}_m)\Big)^{-1}\boldsymbol{e}_1\beta.$$

The corresponding *harmonic Arnoldi (HmArnoldi) approximation* to $f(\boldsymbol{A})\boldsymbol{b}$ is

$$\boldsymbol{f}_m^{\mathrm{HmArn}} = \boldsymbol{B}_m f\Big(\big((\boldsymbol{A}\boldsymbol{B}_m)^*\boldsymbol{B}_m\big)^{-1}(\boldsymbol{A}\boldsymbol{B}_m)^*(\boldsymbol{A}\boldsymbol{B}_m)\Big)\boldsymbol{e}_1\beta, \tag{2.7}$$

provided that the corresponding matrix function is well-defined. If $\boldsymbol{b}_{m+1} \perp \boldsymbol{A}\boldsymbol{B}_m$ in the Arnoldi-like decomposition (2.4), then

$$\big((\boldsymbol{A}\boldsymbol{B}_m)^*\boldsymbol{B}_m\big)^{-1}(\boldsymbol{A}\boldsymbol{B}_m)^*(\boldsymbol{A}\boldsymbol{B}_m) = \boldsymbol{H}_m \quad \text{and} \quad \boldsymbol{f}_m^{\mathrm{HmArn}} = \boldsymbol{B}_m f(\boldsymbol{H}_m)\boldsymbol{e}_1\beta. \tag{2.8}$$

Thus, under this orthogonality condition, the harmonic Arnoldi approximation has the same form as the Arnoldi approximation in (2.6). For $t = 0$, $\boldsymbol{x}_m^{\mathrm{HmArn}}(0)$ coincides with the generalized minimal residual (GMRES) approximation to the solution of (2.3) with a zero initial guess [21]. For general shifts t, harmonic Arnoldi retains the same test space $\boldsymbol{A}\mathcal{K}_m(\boldsymbol{A},\boldsymbol{b})$. This connection to GMRES underlies the convergence analysis of restarted harmonic Arnoldi for Stieltjes functions of positive real matrices [11, Section 6]. These convergence results motivate our consideration of harmonic Arnoldi and its sketched extension.

2.2. Sketched Krylov subspace methods. In the Krylov subspace methods introduced in Section 2.1, the shifted linear system (2.3) is approximated by imposing a Petrov–Galerkin condition on the residual. This procedure can be accelerated using the so-called *sketch-and-solve* approach [23]. More specifically, suppose that, for a fixed constant $\varepsilon \in [0,1)$, the matrix $\boldsymbol{S} \in \mathbb{C}^{s\times N}$ satisfies the following *subspace embedding property*:

$$(1-\varepsilon)\|\boldsymbol{v}\|^2 \le \|\boldsymbol{S}\boldsymbol{v}\|^2 \le (1+\varepsilon)\|\boldsymbol{v}\|^2, \qquad \forall \boldsymbol{v} \in \mathcal{K}_{m+1}(\boldsymbol{A},\boldsymbol{b}). \tag{2.9}$$

Condition (2.9) admits the following equivalent bound for the inner product:

$$\big|\langle \boldsymbol{u},\boldsymbol{v}\rangle - \langle \boldsymbol{S}\boldsymbol{u},\boldsymbol{S}\boldsymbol{v}\rangle\big| \le \varepsilon\|\boldsymbol{u}\|\|\boldsymbol{v}\|, \qquad \forall \boldsymbol{u},\boldsymbol{v} \in \mathcal{K}_{m+1}(\boldsymbol{A},\boldsymbol{b}). \tag{2.10}$$

For completeness, we briefly verify this equivalence for complex vectors with the same ε.

Let $\boldsymbol{V}$ be an orthonormal basis of $\mathcal{K}_{m+1}(\boldsymbol{A}, \boldsymbol{b})$, and define $\boldsymbol{E} = \boldsymbol{V}^*(\boldsymbol{I} - \boldsymbol{S}^*\boldsymbol{S})\boldsymbol{V}$. Since $\boldsymbol{E}$ is Hermitian, (2.9) gives

$$\|\boldsymbol{E}\| = \max_{\|\boldsymbol{z}\|=1} \left|\boldsymbol{z}^* \boldsymbol{E} \boldsymbol{z}\right| \le \varepsilon.$$

Writing $\boldsymbol{u} = \boldsymbol{V}\boldsymbol{x}$ and $\boldsymbol{v} = \boldsymbol{V}\boldsymbol{y}$, we obtain

$$\left|\langle \boldsymbol{u}, \boldsymbol{v}\rangle - \langle \boldsymbol{S}\boldsymbol{u}, \boldsymbol{S}\boldsymbol{v}\rangle\right| = \left|\boldsymbol{x}^* \boldsymbol{E} \boldsymbol{y}\right| \le \|\boldsymbol{E}\|\|\boldsymbol{x}\|\|\boldsymbol{y}\| \le \varepsilon \|\boldsymbol{u}\|\|\boldsymbol{v}\|.$$

Conversely, taking $\boldsymbol{u} = \boldsymbol{v}$ in (2.10) recovers (2.9).

We call such a matrix $\boldsymbol{S}$ a *sketching matrix*, and s the sketching dimension. Two vectors $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{C}^N$ are called $\boldsymbol{S}$-orthogonal, denoted by $\boldsymbol{x} \perp_{\boldsymbol{S}} \boldsymbol{y}$, if $\boldsymbol{S}\boldsymbol{x} \perp \boldsymbol{S}\boldsymbol{y}$. Typically, $\boldsymbol{S}$ is drawn from an *oblivious subspace embedding* (OSE) family. For $\varepsilon, \delta_{\text{fail}} \in (0,1)$, an $(\varepsilon, \delta_{\text{fail}})$ OSE family is a distribution over matrices $\boldsymbol{S} \in \mathbb{C}^{s\times N}$ that is independent of the target subspace and has the following property: for every fixed subspace $\mathcal{U} \subseteq \mathbb{C}^N$ with $\dim \mathcal{U} \le m+1$, the norm inequalities in (2.9) hold simultaneously for all $\boldsymbol{v} \in \mathcal{U}$ with probability at least $1 - \delta_{\text{fail}}$ [23]. For example, consider sparse sign matrices with independent columns,

$$\boldsymbol{S} = \frac{1}{\sqrt{\zeta}} \begin{bmatrix} \boldsymbol{s}_1 & \cdots & \boldsymbol{s}_N \end{bmatrix} \in \mathbb{R}^{s\times N}.$$

The columns are drawn independently. Each column $\boldsymbol{s}_i$ has exactly ζ nonzero entries, whose positions are selected uniformly without replacement from the s available positions. The nonzero entries are assigned independent values ± 1 with equal probability. For any $B > 2$, this family provides the above guarantee whenever

$$s \ge C_1 \frac{B(m+1)\log((m+1)/\delta_{\text{fail}})}{\varepsilon^2}, \qquad \zeta \ge C_2 \frac{\log((m+1)/\delta_{\text{fail}})}{\varepsilon \log B},$$

where $C_1, C_2 > 0$ are absolute constants and $1 \le \zeta \le s$ [5].

We first consider the sketched version of the Arnoldi approximation. Let $\boldsymbol{x}_m^{\text{sArn}}(t)$ approximate the solution $\boldsymbol{x}(t)$ of (2.3). The Petrov–Galerkin condition is imposed after sketching, namely, $\boldsymbol{r}_m \perp_{\boldsymbol{S}} \mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b})$. By the subspace embedding property (2.9), $\boldsymbol{S}\boldsymbol{B}_m$ has full column rank. Following the same argument as in Section 2.1, we obtain

$$\boldsymbol{y}_m^{\text{sArn}}(t) = \big(t\boldsymbol{I}_m - (\boldsymbol{S}\boldsymbol{B}_m)^\dagger (\boldsymbol{S}\boldsymbol{A}\boldsymbol{B}_m)\big)^{-1} \boldsymbol{e}_1 \beta$$

and

$$\boldsymbol{f}_m^{\text{sArn}} = \boldsymbol{B}_m f\big((\boldsymbol{S}\boldsymbol{B}_m)^\dagger (\boldsymbol{S}\boldsymbol{A}\boldsymbol{B}_m)\big)\boldsymbol{e}_1 \beta, \tag{2.11}$$

provided that the corresponding matrix function is well-defined. The vector $\boldsymbol{f}_m^{\text{sArn}}$ in (2.11) is called the *sketched Arnoldi (sArnoldi) approximation* to $f(\boldsymbol{A})\boldsymbol{b}$. We point out that the sArnoldi approximation (2.11) used here differs in form from that in [15], although the two formulations are mathematically equivalent. Moreover, if $\boldsymbol{b}_{m+1} \perp_{\boldsymbol{S}} \boldsymbol{B}_m$ in the Arnoldi-like decomposition (2.4), then

$$(\boldsymbol{S}\boldsymbol{B}_m)^\dagger (\boldsymbol{S}\boldsymbol{A}\boldsymbol{B}_m) = \boldsymbol{H}_m \quad \text{and} \quad \boldsymbol{f}_m^{\text{sArn}} = \boldsymbol{B}_m f(\boldsymbol{H}_m)\boldsymbol{e}_1 \beta. \tag{2.12}$$

A similar discussion applies to the HmArnoldi approximation. By imposing the condition $\boldsymbol{r}_m \perp_{\boldsymbol{S}} \boldsymbol{A}\mathcal{K}_m(\boldsymbol{A},\boldsymbol{b})$, we obtain

$$\boldsymbol{y}_m^{\mathrm{sHmArn}}(t) = \Big(t\boldsymbol{I}_m - \big((\boldsymbol{SAB}_m)^*(\boldsymbol{SB}_m)\big)^{-1}(\boldsymbol{SAB}_m)^*(\boldsymbol{SAB}_m)\Big)^{-1}\boldsymbol{e}_1\beta.$$

Consequently, $\boldsymbol{x}_m^{\mathrm{sHmArn}}(t) = \boldsymbol{B}_m\boldsymbol{y}_m^{\mathrm{sHmArn}}(t)$ is the sketched harmonic Arnoldi approximation to the solution of the shifted linear system (2.3). The corresponding *sketched harmonic Arnoldi (sHmArnoldi) approximation* to $f(\boldsymbol{A})\boldsymbol{b}$ is

$$\boldsymbol{f}_m^{\mathrm{sHmArn}} = \boldsymbol{B}_m f\Big(\big((\boldsymbol{SAB}_m)^*(\boldsymbol{SB}_m)\big)^{-1}(\boldsymbol{SAB}_m)^*(\boldsymbol{SAB}_m)\Big)\boldsymbol{e}_1\beta, \tag{2.13}$$

provided that the corresponding matrix function is well-defined. If $\boldsymbol{b}_{m+1} \perp_{\boldsymbol{S}} \boldsymbol{AB}_m$ in the Arnoldi-like decomposition (2.4), then

$$\begin{aligned}\big((\boldsymbol{SAB}_m)^*(\boldsymbol{SB}_m)\big)^{-1}(\boldsymbol{SAB}_m)^*(\boldsymbol{SAB}_m) &= \boldsymbol{H}_m,\\ \boldsymbol{f}_m^{\mathrm{sHmArn}} &= \boldsymbol{B}_m f(\boldsymbol{H}_m)\boldsymbol{e}_1\beta.\end{aligned} \tag{2.14}$$

2.3. Rank-1 update of the Arnoldi-like decomposition. The discussions in Sections 2.1 and 2.2 show that enforcing a suitable orthogonality condition on the last vector $\boldsymbol{b}_{m+1}$ can simplify the approximation formula by reducing it to a matrix function of the upper Hessenberg matrix $\boldsymbol{H}_m$. This condition is not always satisfied by a given Arnoldi-like decomposition. It can, however, be enforced by applying a rank-1 update to (2.4), as described below; see also [6, 13] for related approaches. The update below is a specialization of Stewart's translation operation for Krylov decompositions [22].

Given (2.4), we update $\boldsymbol{b}_{m+1}$ to $\widetilde{\boldsymbol{b}}_{m+1}$ so that $\widetilde{\boldsymbol{b}}_{m+1}$ satisfies a prescribed orthogonality condition. To this end, write $\boldsymbol{b}_{m+1} = \boldsymbol{B}_m\boldsymbol{c}_m + \widetilde{\boldsymbol{b}}_{m+1}\alpha_{m+1}$, where $\boldsymbol{c}_m$ is to be determined and α_{m+1} is chosen according to a normalization condition. Substituting this expression into (2.4) gives

$$\begin{aligned}\boldsymbol{AB}_m &= \boldsymbol{B}_m\boldsymbol{H}_m + \big(\boldsymbol{B}_m\boldsymbol{c}_m + \widetilde{\boldsymbol{b}}_{m+1}\alpha_{m+1}\big)h_{m+1,m}\boldsymbol{e}_m^\top\\ &= \boldsymbol{B}_m\big(\boldsymbol{H}_m + \boldsymbol{c}_m h_{m+1,m}\boldsymbol{e}_m^\top\big) + \widetilde{\boldsymbol{b}}_{m+1}\big(\alpha_{m+1}h_{m+1,m}\big)\boldsymbol{e}_m^\top\\ &= \boldsymbol{B}_m\widetilde{\boldsymbol{H}}_m + \widetilde{\boldsymbol{b}}_{m+1}\tilde{h}_{m+1,m}\boldsymbol{e}_m^\top,\end{aligned}$$

where $\widetilde{\boldsymbol{H}}_m = \boldsymbol{H}_m + \boldsymbol{c}_m h_{m+1,m}\boldsymbol{e}_m^\top$ and $\tilde{h}_{m+1,m} = \alpha_{m+1}h_{m+1,m}$. The matrix $\widetilde{\boldsymbol{H}}_m$ remains upper Hessenberg.

If the desired orthogonality condition is $\widetilde{\boldsymbol{b}}_{m+1} \perp \boldsymbol{B}_m$ or $\widetilde{\boldsymbol{b}}_{m+1} \perp_{\boldsymbol{S}} \boldsymbol{B}_m$, then $\boldsymbol{c}_m = \boldsymbol{B}_m^\dagger\boldsymbol{b}_{m+1}$ or $\boldsymbol{c}_m = (\boldsymbol{SB}_m)^\dagger(\boldsymbol{S}\boldsymbol{b}_{m+1})$, respectively. Similarly, if the desired orthogonality condition is $\widetilde{\boldsymbol{b}}_{m+1} \perp \boldsymbol{AB}_m$ or $\widetilde{\boldsymbol{b}}_{m+1} \perp_{\boldsymbol{S}} \boldsymbol{AB}_m$, then

$$\begin{aligned}\boldsymbol{c}_m &= \big((\boldsymbol{AB}_m)^*\boldsymbol{B}_m\big)^{-1}(\boldsymbol{AB}_m)^*\boldsymbol{b}_{m+1}\\ \text{or}\quad \boldsymbol{c}_m &= \big((\boldsymbol{SAB}_m)^*(\boldsymbol{SB}_m)\big)^{-1}(\boldsymbol{SAB}_m)^*(\boldsymbol{S}\boldsymbol{b}_{m+1}),\end{aligned}$$

respectively. The Euclidean orthogonality condition $\widetilde{\boldsymbol{b}}_{m+1} \perp \boldsymbol{B}_m$ and the corresponding rank-1 correction are also used in the similarity-restoring randomized Arnoldi process of [13].

Under the respective orthogonality conditions $\boldsymbol{b}_{m+1} \perp \boldsymbol{B}_m$, $\boldsymbol{b}_{m+1} \perp \boldsymbol{AB}_m$, $\boldsymbol{b}_{m+1} \perp_{\boldsymbol{S}} \boldsymbol{B}_m$, and $\boldsymbol{b}_{m+1} \perp_{\boldsymbol{S}} \boldsymbol{AB}_m$, the Arnoldi, HmArnoldi, sArnoldi, and sHmArnoldi

approximations take the common *Arnoldi-like approximation* form

$$\boldsymbol{f}_m \coloneqq \boldsymbol{B}_m f(\boldsymbol{H}_m)\boldsymbol{e}_1\beta. \tag{2.15}$$

Here, $\boldsymbol{B}_m$, $\boldsymbol{H}_m$, and β are method-dependent.

3. Restarting based on quadrature. This section formulates the integral error representation and quadrature-based restarting of [12] for the four approximations in Section 2, retaining the normalization factors for each cycle. Our algorithmic contributions in Section 4 combine this mechanism with sketched basis generation, including the sketched Arnoldi process, the new sketched harmonic Arnoldi process, and adaptive truncated Arnoldi processes. The following lemma recalls the required integral representation of the interpolating polynomial [12, Lemma 3.1].

LEMMA 3.1 (Integral representation of the interpolating polynomial). *Let $\Omega \subset \mathbb{C}$ be a domain, and let $f\colon \Omega \to \mathbb{C}$ be analytic on the closure of Ω and admit the integral representation* (2.1)*. Then the polynomial $p_{m-1} \in \mathcal{P}_{m-1}$ interpolating f at the points $\{\theta_1, \dots, \theta_m\} \subset \Omega$ is given by*

$$p_{m-1}(z) = \int_\Gamma \left(1 - \frac{\phi_m(z)}{\phi_m(t)}\right) \frac{g(t)}{t-z}\,\mathrm{d}t, \tag{3.1}$$

where $\phi_m(z) = (z-\theta_1)\cdots(z-\theta_m)$, provided that the integral in (3.1) *exists.*

3.1. Restarting the Arnoldi-like approximation. We first recall the following polynomial identity for Arnoldi-like decompositions [20, Lemma 2.1].

LEMMA 3.2 (Polynomial action in an Arnoldi-like decomposition). *Given an Arnoldi-like decomposition* (2.4)*, let $p(z) = c_0 + c_1 z + \cdots + c_m z^m \in \mathcal{P}_m$. Then*

$$p(\boldsymbol{A})\boldsymbol{b}_1 = \boldsymbol{B}_m p(\boldsymbol{H}_m)\boldsymbol{e}_1 + \boldsymbol{b}_{m+1}\gamma_m c_m,$$

where $\gamma_m = \prod_{j=1}^m h_{j+1,j}$.

For the Arnoldi-like decomposition (2.4), consider its corresponding Arnoldi-like approximation $\boldsymbol{f}_m$ in (2.15). Assume that $\Lambda(\boldsymbol{A}) \cup \Lambda(\boldsymbol{H}_m) \subset \Omega$, and that f is analytic on the closure of Ω and admits (2.1). Let $p_{m-1} \in \mathcal{P}_{m-1}$ interpolate f at $\Lambda(\boldsymbol{H}_m)$. Then $f(\boldsymbol{H}_m) = p_{m-1}(\boldsymbol{H}_m)$. Since its degree is at most $m-1$, $c_m = 0$ in Lemma 3.2; multiplying that identity by β gives

$$\boldsymbol{f}_m = \boldsymbol{B}_m p_{m-1}(\boldsymbol{H}_m)\boldsymbol{e}_1\beta = p_{m-1}(\boldsymbol{A})\boldsymbol{b}_1\beta = p_{m-1}(\boldsymbol{A})\boldsymbol{b},$$

where we used $\boldsymbol{b} = \boldsymbol{b}_1\beta$. Write $\Lambda(\boldsymbol{H}_m) = \{\theta_1, \dots, \theta_m\}$ and define $\phi_m(z) = (z - \theta_1)\cdots(z-\theta_m)$. Then Lemma 3.1 gives

$$f(z) - p_{m-1}(z) = \int_\Gamma \frac{\phi_m(z)}{\phi_m(t)} \frac{g(t)}{t-z}\,\mathrm{d}t,$$

and therefore

$$f(\boldsymbol{A})\boldsymbol{b} - \boldsymbol{f}_m = \left(\int_\Gamma \frac{g(t)}{\phi_m(t)}(t\boldsymbol{I} - \boldsymbol{A})^{-1}\,\mathrm{d}t\right)\phi_m(\boldsymbol{A})\boldsymbol{b}.$$

Moreover, Lemma 3.2 gives

$$\phi_m(\boldsymbol{A})\boldsymbol{b} = \boldsymbol{B}_m\phi_m(\boldsymbol{H}_m)\boldsymbol{e}_1\beta + \boldsymbol{b}_{m+1}\gamma_m\beta = \boldsymbol{b}_{m+1}\gamma_m\beta,$$

where we used $\phi_m(\boldsymbol{H}_m) = \boldsymbol{0}$. Thus,

$$f(\boldsymbol{A})\boldsymbol{b} - \boldsymbol{f}_m = \gamma_m\beta\Big(\int_\Gamma \frac{g(t)}{\phi_m(t)}(t\boldsymbol{I} - \boldsymbol{A})^{-1}\,\mathrm{d}t\Big)\boldsymbol{b}_{m+1}.$$

The following theorem states the integral error formula of [12, Theorem 3.4] in the Arnoldi-like formulation used here, with the normalization factor β retained explicitly.

THEOREM 3.3 (Error of the Arnoldi-like approximation). *Consider the Arnoldi-like decomposition* (2.4) *with* $\boldsymbol{b} = \boldsymbol{b}_1\beta$. *Suppose that* f *admits the integral representation* (2.1), *and let* $\boldsymbol{f}_m$ *be the Arnoldi-like approximation defined in* (2.15). *Define* $\phi_m(t) = (t-\theta_1)\cdots(t-\theta_m)$, *where* $\Lambda(\boldsymbol{H}_m) = \{\theta_1,\dots,\theta_m\} \subset \Omega$, *and let* $\gamma_m = \prod_{j=1}^m h_{j+1,j}$. *Then*

$$f(\boldsymbol{A})\boldsymbol{b} - \boldsymbol{f}_m = \gamma_m\beta\Big(\int_\Gamma \frac{g(t)}{\phi_m(t)}(t\boldsymbol{I} - \boldsymbol{A})^{-1}\,\mathrm{d}t\Big)\boldsymbol{b}_{m+1} =: \mathrm{err}(\boldsymbol{A})\boldsymbol{b}_{m+1}, \tag{3.2}$$

provided that the integral exists. Thus, the error function is

$$\mathrm{err}(z) = \gamma_m\beta\int_\Gamma \frac{g(t)}{\phi_m(t)}(t-z)^{-1}\,\mathrm{d}t.$$

The choice of $\boldsymbol{b}_{m+1}$ as the restart vector follows directly from the error representation (3.2). Using $\boldsymbol{b}_{m+1}$ as the new starting vector, suppose that a new Arnoldi-like decomposition $\boldsymbol{A}\boldsymbol{B}_m^+ = \boldsymbol{B}_{m+1}^+\underline{\boldsymbol{H}}_m^+$ is generated with $\boldsymbol{b}_{m+1} = \boldsymbol{b}_1^+\beta^+$. Applying Theorem 3.3 to err and the new decomposition under its stated assumptions gives

$$\mathrm{err}(\boldsymbol{A})\boldsymbol{b}_{m+1} = \boldsymbol{B}_m^+\mathrm{err}(\boldsymbol{H}_m^+)\boldsymbol{e}_1\beta^+ + \mathrm{err}^+(\boldsymbol{A})\boldsymbol{b}_{m+1}^+,$$

where $\mathrm{err}^+(z)$ is the new error function. The first term is the Arnoldi-like approximation and the second its exact remaining error.

Algorithm 3.1 repeats this process. For $k \geq 1$, it determines $\mathrm{err}^{[k]}$ from the preceding decompositions using Corollary 3.4. The new decomposition starts from $\boldsymbol{b}_{m+1}^{[k-1]} = \boldsymbol{b}_1^{[k]}\beta^{[k]}$. Adaptive quadrature evaluates $\mathrm{err}^{[k]}(\boldsymbol{H}_m^{[k]})\boldsymbol{e}_1\beta^{[k]}$ using shifted systems with the current reduced matrix. Multiplying the result by $\boldsymbol{B}_m^{[k]}$ gives the correction $\boldsymbol{g}^{[k]}$, which is added to the current approximation. Although the corollary assumes a fixed Krylov subspace dimension m in each restart cycle, the construction extends directly to variable dimensions.

COROLLARY 3.4 (Error of the Arnoldi-like approximation after k restarts). *Under the assumptions of Theorem 3.3, suppose that the* ℓ*-th cycle generates an Arnoldi-like decomposition* $\boldsymbol{A}\boldsymbol{B}_m^{[\ell]} = \boldsymbol{B}_{m+1}^{[\ell]}\underline{\boldsymbol{H}}_m^{[\ell]}$ *for* $0 \leq \ell \leq k$. *Here,* $\boldsymbol{b} = \boldsymbol{b}_1^{[0]}\beta^{[0]}$ *and* $\boldsymbol{b}_{m+1}^{[\ell-1]} = \boldsymbol{b}_1^{[\ell]}\beta^{[\ell]}$ *for* $1 \leq \ell \leq k$. *Assume that* $\Lambda(\boldsymbol{H}_m^{[\ell]}) = \{\theta_j^{[\ell]}\}_{j=1}^m \subset \Omega$, *and define* $\phi_m^{[\ell]}(t) = \prod_{j=1}^m(t-\theta_j^{[\ell]})$ *and* $\gamma_m^{[\ell]} = \prod_{j=1}^m h_{j+1,j}^{[\ell]}$ *for* $0 \leq \ell \leq k$. *Then the error after* k *restarts is*

$$\begin{aligned} f(\boldsymbol{A})\boldsymbol{b} - \boldsymbol{f}^{[k]} &= \Big(\prod_{\ell=0}^k \gamma_m^{[\ell]}\beta^{[\ell]}\Big)\Big(\int_\Gamma \frac{g(t)}{\prod_{\ell=0}^k \phi_m^{[\ell]}(t)}(t\boldsymbol{I} - \boldsymbol{A})^{-1}\,\mathrm{d}t\Big)\boldsymbol{b}_{m+1}^{[k]} \\ &=: \mathrm{err}^{[k+1]}(\boldsymbol{A})\boldsymbol{b}_{m+1}^{[k]}. \end{aligned} \tag{3.3}$$

Following the quadrature approach of [12], we approximate the reduced error

Algorithm 3.1 Restarting framework for computing $f(\boldsymbol{A})\boldsymbol{b}$

Input: Matrix $\boldsymbol{A}$, vector $\boldsymbol{b}$, function f, subspace dimension m
Output: $\boldsymbol{f}^{[k]} \approx f(\boldsymbol{A})\boldsymbol{b}$

1: Compute an Arnoldi-like decomposition $\boldsymbol{A}\boldsymbol{B}_m^{[0]} = \boldsymbol{B}_{m+1}^{[0]}\underline{\boldsymbol{H}}_m^{[0]}$ with $\boldsymbol{b} = \boldsymbol{b}_1^{[0]}\beta^{[0]}$
2: $\boldsymbol{f}^{[0]} = \boldsymbol{B}_m^{[0]} f(\boldsymbol{H}_m^{[0]})\boldsymbol{e}_1\beta^{[0]}$
3: **for** $k = 1, 2, \ldots$ **do**
4: Determine the error function $\mathrm{err}^{[k]}(z)$ using Corollary 3.4
5: Compute an Arnoldi-like decomposition $\boldsymbol{A}\boldsymbol{B}_m^{[k]} = \boldsymbol{B}_{m+1}^{[k]}\underline{\boldsymbol{H}}_m^{[k]}$ with $\boldsymbol{b}_{m+1}^{[k-1]} = \boldsymbol{b}_1^{[k]}\beta^{[k]}$
6: Use adaptive quadrature to compute $\boldsymbol{g}^{[k]} \approx \boldsymbol{B}_m^{[k]}\mathrm{err}^{[k]}(\boldsymbol{H}_m^{[k]})\boldsymbol{e}_1\beta^{[k]}$
7: Update $\boldsymbol{f}^{[k]} = \boldsymbol{f}^{[k-1]} + \boldsymbol{g}^{[k]}$
8: **end for**

vector by

$$\mathrm{err}^{[k]}(\boldsymbol{H}_m^{[k]})\boldsymbol{e}_1\beta^{[k]} \approx \left(\prod_{\ell=0}^{k-1}\gamma_m^{[\ell]}\beta^{[\ell]}\right)\sum_{j=1}^{r}\frac{g(t_j)}{\prod_{\ell=0}^{k-1}\phi_m^{[\ell]}(t_j)}\omega_j^{[k]}(t_j\boldsymbol{I}_m - \boldsymbol{H}_m^{[k]})^{-1}\boldsymbol{e}_1\beta^{[k]},$$

where t_j are quadrature nodes and $\omega_j^{[k]}$ are weights. In practice, the error functions are evaluated using adaptive quadrature. For a given error function, we compute two quadrature approximations using r_1 and r_2 nodes, where $r_2 = 2r_1$. We accept the result with r_2 nodes if the reduced vectors differ by less than the prescribed tolerance times its norm. Otherwise, we set $r_1 = r_2$, double r_2, and repeat the test until the tolerance is satisfied or a prescribed maximum number of nodes is reached. At each subsequent restart cycle, we try halving r_1, subject to the minimum node count, set $r_2 = 2r_1$, and increase the node counts if necessary.

3.2. Restarting Krylov and sketched Krylov subspace methods. Under the orthogonality conditions specified in Section 2, the four Krylov and sketched Krylov approximations share the error representation and quadrature-based restarting formula derived above. This observation forms the basis of the sketch-and-restart framework developed in the next section.

4. Sketch-and-restart algorithms for matrix functions. The previous section showed that quadrature-based restarting applies to any Arnoldi-like approximation of the form $\boldsymbol{B}_m f(\boldsymbol{H}_m)\boldsymbol{e}_1\beta$, provided that the required orthogonality condition is satisfied. In this section, we instantiate this restarting mechanism using different strategies for generating Arnoldi-like decompositions. Specifically, we consider three such strategies: the existing sketched Arnoldi process (Algorithm 4.1), a new sketched harmonic Arnoldi process introduced in this work (Algorithm 4.2), and an adaptive truncated Arnoldi process combined with a rank-1 update (Algorithm 4.3). The first two are used in the fixed-dimensional restarted method (Algorithm 4.4), while the third is used in the adaptive restarted method (Algorithm 4.5). Table 1 summarizes the restarted methods.

4.1. Basis generation. The standard Arnoldi process [21] generates an orthonormal basis for the Krylov subspace $\mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b})$, but its orthogonalization and storage costs can become substantial when m is large. In the sketch-and-restart framework, full orthogonality is not required. It suffices to construct an Arnoldi-like

Table 1: Overview of the restarted methods, all returning $\boldsymbol{f}^{[k]} \approx f(\boldsymbol{A})\boldsymbol{b}$.

Source	Method	Dimension
[12]	Restarted Arnoldi method	Fixed
[11]	Restarted harmonic Arnoldi method	Fixed
Algorithm 4.4	Restarted sketched Arnoldi or restarted sketched harmonic Arnoldi methods	Fixed
Algorithm 4.5	Restarted adaptive truncated Arnoldi method with rank-1 update	Adaptive

decomposition that satisfies the orthogonality condition required by the corresponding approximation. We therefore consider three basis generation strategies that produce possibly nonorthonormal bases with different computational and storage trade-offs.

The first strategy is the *sketched Arnoldi (s-Arnoldi) process* [2, 8]. It is based on the Arnoldi process but computes the Gram–Schmidt coefficients in the sketched space. The same coefficients are then used to update the full-space basis vectors, as summarized in Algorithm 4.1. We use a modified Gram–Schmidt formulation in Algorithm 4.1 and assume that no breakdown occurs for simplicity. In exact arithmetic, these paired updates preserve $\boldsymbol{p}_{j+1} \perp \boldsymbol{P}_j$ and $\boldsymbol{p}_{j+1} = \boldsymbol{S}\boldsymbol{b}_{j+1}$. The algorithm yields a standard Arnoldi decomposition in the sketched space, and the resulting Arnoldi-like decomposition satisfies $\boldsymbol{b}_{m+1} \perp_{\boldsymbol{S}} \boldsymbol{B}_m$. Therefore, the sArnoldi approximation can be computed directly from (2.12). In floating-point arithmetic, however, sketch orthogonality may deteriorate and the stored $\boldsymbol{p}_{j+1}$ may differ from $\boldsymbol{S}\boldsymbol{b}_{j+1}$. Reorthogonalization and recomputing the sketch are possible stabilization measures [2, 8]. The complexity of the s-Arnoldi process under complex arithmetic is $mT_{\text{apply}} + mT_{\text{sketching}} + 4Nm^2 + \mathcal{O}(sm^2 + (N+s)m)$, where T_{apply} and $T_{\text{sketching}}$ denote the costs of applying $\boldsymbol{A}$ and $\boldsymbol{S}$ to a vector, respectively. For fixed m, when the application of $\boldsymbol{A}$ and $\boldsymbol{S}$ are cheap, and the computation in the sketched space is inexpensive, the dominant cost is $4Nm^2$. Consequently, s-Arnoldi nearly halves the orthogonalization cost of the standard Arnoldi process, which is $8Nm^2$.

The second strategy is the *sketched harmonic Arnoldi (sHm-Arnoldi) process*. In [11, Section 6], the reduced matrix $\boldsymbol{H}_m$ is constructed by the standard Arnoldi process followed by a rank-1 update. In constrast, our new proposed sketched harmonic Arnoldi process constructs $\boldsymbol{H}_m$ directly, without requiring a rank-1 update. Its non-sketched counterpart, the *harmonic Arnoldi process*, which is also first proposed in this work, constructs $\boldsymbol{H}_m$ directly as well. Algorithm 4.2 summarizes the sHm-Arnoldi process, where $\{\boldsymbol{b}_1, \ldots, \boldsymbol{b}_j\}$ and $\{\boldsymbol{z}_1, \ldots, \boldsymbol{z}_j\}$ denote the bases of $\mathcal{K}_j(\boldsymbol{A}, \boldsymbol{b})$ and $\mathcal{K}_j(\boldsymbol{A}, \boldsymbol{A}\boldsymbol{b})$, respectively, and $\{\boldsymbol{p}_1, \ldots, \boldsymbol{p}_j\}$ and $\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_j\}$ denote their sketches. As in Algorithm 4.1, Algorithm 4.2 uses sequential paired updates in the sketched and full spaces, and we assume that no breakdown occurs for simplicity. The scaled images $\boldsymbol{z}_i = \boldsymbol{A}\boldsymbol{b}_i/\rho_i^*$ and their sketches $\boldsymbol{q}_i = \boldsymbol{S}\boldsymbol{z}_i$ satisfy $\langle \boldsymbol{p}_i, \boldsymbol{q}_i \rangle = 1$ in exact arithmetic. At each inner iteration, the coefficient is computed from the current sketched vector $\widetilde{\boldsymbol{p}}_{j+1}$ using $\boldsymbol{q}_i$ as the test vector, and the same coefficient is used to update the sketched and full-space vectors $\boldsymbol{p}_{j+1}$ and $\boldsymbol{b}_{j+1}$. In exact arithmetic, these paired updates preserve $\boldsymbol{p}_{j+1} = \boldsymbol{S}\boldsymbol{b}_{j+1}$ and enforce $\boldsymbol{p}_{j+1} \perp \boldsymbol{Q}_j$.

In particular, Algorithm 4.2 returns a nonorthonormal basis $\boldsymbol{B}_{m+1}$ satisfying

Algorithm 4.1 Sketched Arnoldi (s-Arnoldi) process

Function: $[\beta, \boldsymbol{B}_{m+1}, \boldsymbol{P}_{m+1}, \underline{\boldsymbol{H}}_m] = \texttt{sArnoldi}(\boldsymbol{S}, \boldsymbol{A}, \boldsymbol{b}, \boldsymbol{p}, m)$
Input: Sketching matrix $\boldsymbol{S}$, matrix $\boldsymbol{A}$, vector $\boldsymbol{b}$, sketched vector $\boldsymbol{p} = \boldsymbol{S}\boldsymbol{b}$, subspace dimension m
Output: Arnoldi-like decomposition $\boldsymbol{A}\boldsymbol{B}_m = \boldsymbol{B}_{m+1}\underline{\boldsymbol{H}}_m$ where $\boldsymbol{b} = \boldsymbol{b}_1\beta$ and $\boldsymbol{P}_{m+1} = \boldsymbol{S}\boldsymbol{B}_{m+1}$ is orthonormal
$\widetilde{\boldsymbol{b}}_1 = \boldsymbol{b}$, $\widetilde{\boldsymbol{p}}_1 = \boldsymbol{p}$
$\beta = \|\widetilde{\boldsymbol{p}}_1\|$, $\boldsymbol{p}_1 = \widetilde{\boldsymbol{p}}_1/\beta$, $\boldsymbol{b}_1 = \widetilde{\boldsymbol{b}}_1/\beta$
for $j = 1, 2, \ldots, m$ **do**
  $\widetilde{\boldsymbol{b}}_{j+1} = \boldsymbol{A}\boldsymbol{b}_j$, $\widetilde{\boldsymbol{p}}_{j+1} = \boldsymbol{S}\widetilde{\boldsymbol{b}}_{j+1}$
  for $i = 1, \ldots, j$ **do**
    $h_{i,j} = \langle \widetilde{\boldsymbol{p}}_{j+1}, \boldsymbol{p}_i \rangle$, $\widetilde{\boldsymbol{p}}_{j+1} = \widetilde{\boldsymbol{p}}_{j+1} - \boldsymbol{p}_i h_{i,j}$, $\widetilde{\boldsymbol{b}}_{j+1} = \widetilde{\boldsymbol{b}}_{j+1} - \boldsymbol{b}_i h_{i,j}$
  end for
  $h_{j+1,j} = \|\widetilde{\boldsymbol{p}}_{j+1}\|$, $\boldsymbol{p}_{j+1} = \widetilde{\boldsymbol{p}}_{j+1}/h_{j+1,j}$, $\boldsymbol{b}_{j+1} = \widetilde{\boldsymbol{b}}_{j+1}/h_{j+1,j}$
end for

$\boldsymbol{b}_{j+1} \perp_{\boldsymbol{S}} \boldsymbol{A}\boldsymbol{B}_j$ for $1 \le j \le m$. The resulting decomposition allows the sHmArnoldi approximation to be computed directly from (2.14). Setting $\boldsymbol{S} = \boldsymbol{I}_N$ and remove the sketched variables in Algorithm 4.2 gives the harmonic Arnoldi (Hm-Arnoldi) process. To the best of our knowledge, both the harmonic Arnoldi and sketched harmonic Arnoldi processes are new. The term "harmonic" refers to harmonic Ritz projection with test space $\boldsymbol{A}\mathcal{K}_m(\boldsymbol{A}, \boldsymbol{b})$ and target 0 [11, Section 6]; the eigenvalues of $\boldsymbol{H}_m$ are the corresponding sketched harmonic Ritz values.

The complexity of the sHm-Arnoldi process under complex arithmetic is $mT_{\text{apply}} + mT_{\text{sketching}} + 4Nm^2 + \mathcal{O}(sm^2 + (N+s)m)$. Under the same cost assumptions as for the s-Arnoldi process, the dominant cost is $4Nm^2$. Thus, sHm-Arnoldi also nearly halves the orthogonalization cost of the standard Arnoldi process.

Algorithm 4.2 Sketched harmonic Arnoldi (sHm-Arnoldi) process

Function: $[\beta, \boldsymbol{B}_{m+1}, \boldsymbol{P}_{m+1}, \underline{\boldsymbol{H}}_m] = \texttt{sHmArnoldi}(\boldsymbol{S}, \boldsymbol{A}, \boldsymbol{b}, \boldsymbol{p}, m)$
Input: Sketching matrix $\boldsymbol{S}$, matrix $\boldsymbol{A}$, vector $\boldsymbol{b}$, sketched vector $\boldsymbol{p} = \boldsymbol{S}\boldsymbol{b}$, subspace dimension m
Output: Arnoldi-like decomposition $\boldsymbol{A}\boldsymbol{B}_m = \boldsymbol{B}_{m+1}\underline{\boldsymbol{H}}_m$ where $\boldsymbol{b} = \boldsymbol{b}_1\beta$, $\boldsymbol{P}_{m+1} = \boldsymbol{S}\boldsymbol{B}_{m+1}$, and $\boldsymbol{b}_{j+1} \perp_{\boldsymbol{S}} \boldsymbol{A}\boldsymbol{B}_j$ for $1 \le j \le m$
$\widetilde{\boldsymbol{b}}_1 = \boldsymbol{b}$, $\widetilde{\boldsymbol{p}}_1 = \boldsymbol{p}$
$\beta = \|\widetilde{\boldsymbol{p}}_1\|$, $\boldsymbol{p}_1 = \widetilde{\boldsymbol{p}}_1/\beta$, $\boldsymbol{b}_1 = \widetilde{\boldsymbol{b}}_1/\beta$
for $j = 1, 2, \ldots, m$ **do**
  $\widetilde{\boldsymbol{b}}_{j+1} = \boldsymbol{A}\boldsymbol{b}_j$, $\widetilde{\boldsymbol{p}}_{j+1} = \boldsymbol{S}\widetilde{\boldsymbol{b}}_{j+1}$
  $\rho_j = \langle \boldsymbol{p}_j, \widetilde{\boldsymbol{p}}_{j+1} \rangle$
  $\boldsymbol{z}_j = \widetilde{\boldsymbol{b}}_{j+1}/\rho_j^*$, $\boldsymbol{q}_j = \widetilde{\boldsymbol{p}}_{j+1}/\rho_j^*$
  for $i = 1, \ldots, j$ **do**
    $h_{i,j} = \langle \widetilde{\boldsymbol{p}}_{j+1}, \boldsymbol{q}_i \rangle$, $\widetilde{\boldsymbol{p}}_{j+1} = \widetilde{\boldsymbol{p}}_{j+1} - \boldsymbol{p}_i h_{i,j}$, $\widetilde{\boldsymbol{b}}_{j+1} = \widetilde{\boldsymbol{b}}_{j+1} - \boldsymbol{b}_i h_{i,j}$
  end for
  $h_{j+1,j} = \|\widetilde{\boldsymbol{p}}_{j+1}\|$, $\boldsymbol{p}_{j+1} = \widetilde{\boldsymbol{p}}_{j+1}/h_{j+1,j}$, $\boldsymbol{b}_{j+1} = \widetilde{\boldsymbol{b}}_{j+1}/h_{j+1,j}$
end for

THEOREM 4.1 (Properties of the sHm-Arnoldi process). *Suppose that Algo-*

rithm 4.2 is performed in exact arithmetic, with $\beta \neq 0$, $\rho_j \neq 0$, *and* $h_{j+1,j} \neq 0$ *for* $1 \le j \le m$. *Then,* $\boldsymbol{B}_j$ *is a basis of* $\mathcal{K}_j(\boldsymbol{A}, \boldsymbol{b})$ *for* $1 \le j \le m+1$ *and* $\boldsymbol{Z}_j$ *is a basis of* $\boldsymbol{A}\mathcal{K}_j(\boldsymbol{A}, \boldsymbol{b})$ *for* $1 \le j \le m$. *The generated factors satisfy*

$$\begin{aligned} \boldsymbol{A}\boldsymbol{B}_m &= \boldsymbol{B}_{m+1}\underline{\boldsymbol{H}}_m, & \boldsymbol{b} &= \boldsymbol{b}_1\beta, \\ \boldsymbol{P}_{m+1} &= \boldsymbol{S}\boldsymbol{B}_{m+1}, & \boldsymbol{Q}_m &= \boldsymbol{S}\boldsymbol{Z}_m \end{aligned}$$

and

$$\boldsymbol{b}_{j+1} \perp_{\boldsymbol{S}} \boldsymbol{A}\boldsymbol{B}_j, \qquad 1 \le j \le m.$$

Proof. We prove the result from a projection perspective. The paired updates and scaling give $\boldsymbol{P}_{m+1} = \boldsymbol{S}\boldsymbol{B}_{m+1}$, $\boldsymbol{Q}_m = \boldsymbol{S}\boldsymbol{Z}_m$, $\|\boldsymbol{p}_i\| = 1$ for $1 \le i \le m+1$, and $\langle \boldsymbol{p}_i, \boldsymbol{q}_i \rangle = 1$ for $1 \le i \le m$. Since $\boldsymbol{z}_i = \boldsymbol{A}\boldsymbol{b}_i/\rho_i^*$, the required orthogonality is equivalent to

$$\boldsymbol{b}_{j+1} \perp_{\boldsymbol{S}} \boldsymbol{A}\boldsymbol{B}_j \quad \Longleftrightarrow \quad \boldsymbol{p}_{j+1} \perp \boldsymbol{Q}_j.$$

We proceed by induction, starting from $\boldsymbol{b}_1 = \boldsymbol{b}/\beta$. Suppose that $\boldsymbol{B}_j$ is a basis of $\mathcal{K}_j(\boldsymbol{A}, \boldsymbol{b})$ and $\langle \boldsymbol{p}_i, \boldsymbol{q}_k \rangle = 0$ for $k < i \le j$. Then

$$\boldsymbol{T}_j = \boldsymbol{Q}_j^* \boldsymbol{P}_j = \big(\langle \boldsymbol{p}_k, \boldsymbol{q}_i \rangle\big)_{i,k=1}^{j} \in \mathbb{C}^{j \times j}$$

is unit lower triangular with diagonal entries $\langle \boldsymbol{p}_i, \boldsymbol{q}_i \rangle = 1$. Its invertibility implies that both $\boldsymbol{Q}_j$ and $\boldsymbol{Z}_j$ have full column rank. Since each $\boldsymbol{z}_i$ is a nonzero multiple of $\boldsymbol{A}\boldsymbol{b}_i$, $\boldsymbol{Z}_j$ is therefore a basis of $\boldsymbol{A}\mathcal{K}_j(\boldsymbol{A}, \boldsymbol{b})$.

Define

$$\boldsymbol{\Pi}_j = \boldsymbol{I}_N - \boldsymbol{B}_j \boldsymbol{T}_j^{-1} \boldsymbol{Q}_j^* \boldsymbol{S}.$$

This is an oblique projector satisfying

$$\boldsymbol{\Pi}_j^2 = \boldsymbol{\Pi}_j, \qquad \boldsymbol{\Pi}_j \boldsymbol{B}_j = \boldsymbol{0}, \qquad \boldsymbol{S}\boldsymbol{\Pi}_j = \big(\boldsymbol{I}_s - \boldsymbol{P}_j \boldsymbol{T}_j^{-1} \boldsymbol{Q}_j^*\big)\boldsymbol{S}.$$

In the inner loop, applying the successive sketched updates to any $\boldsymbol{w} \in \mathbb{C}^s$ produces coefficients satisfying

$$c_i = \left\langle \boldsymbol{w} - \sum_{k=1}^{i-1} \boldsymbol{p}_k c_k, \boldsymbol{q}_i \right\rangle, \qquad 1 \le i \le j.$$

Since $\boldsymbol{T}_j$ is unit lower triangular, these are precisely the forward substitution relations for $\boldsymbol{T}_j \boldsymbol{c} = \boldsymbol{Q}_j^* \boldsymbol{w}$, where $\boldsymbol{c} = [c_1, \ldots, c_j]^\top$. Thus, the resulting vector after j updates is

$$\boldsymbol{w} - \boldsymbol{P}_j \boldsymbol{c} = \big(\boldsymbol{I}_s - \boldsymbol{P}_j \boldsymbol{T}_j^{-1} \boldsymbol{Q}_j^*\big)\boldsymbol{w}.$$

As $\boldsymbol{w}$ is arbitrary, this proves

$$(\boldsymbol{I}_s - \boldsymbol{p}_j \boldsymbol{q}_j^*) \cdots (\boldsymbol{I}_s - \boldsymbol{p}_1 \boldsymbol{q}_1^*) = \boldsymbol{I}_s - \boldsymbol{P}_j \boldsymbol{T}_j^{-1} \boldsymbol{Q}_j^*.$$

The paired updates and normalization therefore give

$$\boldsymbol{b}_{j+1} = \frac{\boldsymbol{\Pi}_j \boldsymbol{A}\boldsymbol{b}_j}{h_{j+1,j}}, \qquad \boldsymbol{p}_{j+1} = \frac{\boldsymbol{S}\boldsymbol{\Pi}_j \boldsymbol{A}\boldsymbol{b}_j}{h_{j+1,j}} \perp \boldsymbol{Q}_j,$$

where the last orthogonality follows from

$$\boldsymbol{Q}_j^* \boldsymbol{S} \boldsymbol{\Pi}_j = \boldsymbol{Q}_j^* \big(\boldsymbol{I}_s - \boldsymbol{P}_j \boldsymbol{T}_j^{-1} \boldsymbol{Q}_j^* \big) \boldsymbol{S} = \boldsymbol{0}.$$

Hence, $\boldsymbol{b}_{j+1} \perp_{\boldsymbol{S}} \boldsymbol{A}\boldsymbol{B}_j$.

Since $\|\boldsymbol{p}_{j+1}\| = 1$, $\boldsymbol{b}_{j+1}$ is nonzero, and the idempotence of $\boldsymbol{\Pi}_j$ gives

$$\boldsymbol{\Pi}_j \boldsymbol{b}_{j+1} = \boldsymbol{b}_{j+1} \neq \boldsymbol{0}.$$

Since $\boldsymbol{\Pi}_j \boldsymbol{B}_j = \boldsymbol{0}$, the new vector cannot belong to $\mathcal{K}_j(\boldsymbol{A}, \boldsymbol{b})$. Moreover, the update formula gives $\boldsymbol{b}_{j+1} \in \mathcal{K}_{j+1}(\boldsymbol{A}, \boldsymbol{b})$. Thus, $\boldsymbol{B}_{j+1}$ is a basis of $\mathcal{K}_{j+1}(\boldsymbol{A}, \boldsymbol{b})$, completing the induction. Finally, collecting the recurrence coefficients gives $\boldsymbol{A}\boldsymbol{B}_m = \boldsymbol{B}_{m+1} \underline{\boldsymbol{H}}_m$. ∎

The third strategy is the *truncated Arnoldi (t-Arnoldi) process* [21]. It generates a nonorthonormal basis by orthogonalizing each new vector only against the t most recently generated basis vectors. Let $m_{\mathrm{out}} \leq m_{\max}$ denote the adaptively determined output dimension. For $j < m_{\max}$, whenever the sketching dimension s satisfies $s < \eta(j+1)$, we increase s by s_0, draw a new sketching matrix $\boldsymbol{S} \in \mathbb{C}^{s \times N}$, and recompute $\boldsymbol{P}_{j+1} = \boldsymbol{S}\boldsymbol{B}_{j+1}$. After any such resketching, we check $\kappa(\boldsymbol{P}_{j+1})$ at steps $j = 5, 10, \ldots < m_{\max}$ and stop if it exceeds τ_{cond}. The process also terminates when the dimension reaches $m_{\max}$.

At the end of basis generation, we apply a rank-1 update to the Arnoldi-like decomposition to enforce a prescribed orthogonality condition on the last basis vector; see Section 2.3. When the condition is $\boldsymbol{b}_{m_{\mathrm{out}}+1} \perp \boldsymbol{B}_{m_{\mathrm{out}}}$, we call the resulting method the adaptive Arnoldi (adaArnoldi-t) method. When the condition is $\boldsymbol{b}_{m_{\mathrm{out}}+1} \perp \boldsymbol{A}\boldsymbol{B}_{m_{\mathrm{out}}}$, we call it the adaptive harmonic Arnoldi (adaHmArnoldi-t) method.

The complete procedure is summarized in Algorithm 4.3. As in Algorithms 4.1 and 4.2, we assume that no breakdown occurs for simplicity, including in the final rank-1 update. For a small fixed truncation parameter t, the complexity under complex arithmetic, excluding resketching, is

$$m_{\mathrm{out}} T_{\mathrm{apply}} + m_{\mathrm{out}} T_{\mathrm{sketching}} + 8N m_{\mathrm{out}}^2 + \mathcal{O}((N+s)m_{\mathrm{out}} + m_{\mathrm{out}}^3).$$

Under the same cost assumptions as for the preceding processes, the dominant cost is $8N m_{\mathrm{out}}^2$. Since the resulting Krylov subspace dimension is typically small, the rank-1 update is inexpensive and can be performed at the BLAS-2 level.

4.2. Algorithms. We now present two algorithmic realizations of the sketch-and-restart framework. Algorithm 4.4 uses a fixed Krylov subspace dimension and applies either the s-Arnoldi or the sHm-Arnoldi process in each restart cycle, leading respectively to the restarted sArnoldi and restarted sHmArnoldi methods. The error function is updated as in Algorithm 3.1, and the correction term is evaluated by adaptive quadrature.

Algorithm 4.5 instead chooses the Krylov subspace dimension adaptively using the t-Arnoldi process. The dimension m_0 obtained in the first restart cycle is used as the upper bound $m_{\max}$ in all subsequent cycles, so that the sketching matrix do not need to be redrawn ins subsequent cycles. The same quadrature-based restarting mechanism is then applied to the resulting Arnoldi-like decompositions.

5. Convergence analysis. In this section, we analyze the convergence of the proposed sketch-and-restart algorithms. We assume exact arithmetic and neglect

Algorithm 4.3 Truncated Arnoldi (t-Arnoldi) process

Function: $[\boldsymbol{S}, m_{\text{out}}, \beta, \boldsymbol{B}_{m_{\text{out}}+1}, \boldsymbol{P}_{m_{\text{out}}+1}, \underline{\boldsymbol{H}}_{m_{\text{out}}}]$
$= \texttt{tArnoldi_adaptive}(\boldsymbol{S}, \boldsymbol{A}, \boldsymbol{b}, \boldsymbol{p}, m_{\max}, t, \tau_{\text{cond}}, s_0, \eta, \mathcal{C})$

Input: Sketching matrix $\boldsymbol{S}$, matrix $\boldsymbol{A}$, vector $\boldsymbol{b}$, sketched vector $\boldsymbol{p} = \boldsymbol{S}\boldsymbol{b}$, maximum subspace dimension $m_{\max}$, truncation parameter t, condition number threshold τ_{cond}, sketching dimension increment s_0, sketching dimension control parameter η, and selected orthogonality condition $\mathcal{C}$ for the rank-1 update from Section 2.3

Output: Adaptively determined dimension $m_{\text{out}} \le m_{\max}$, updated sketching matrix $\boldsymbol{S}$, and Arnoldi-like decomposition $\boldsymbol{A}\boldsymbol{B}_{m_{\text{out}}} = \boldsymbol{B}_{m_{\text{out}}+1}\underline{\boldsymbol{H}}_{m_{\text{out}}}$, where $\boldsymbol{b} = \boldsymbol{b}_1\beta$, $\boldsymbol{P}_{m_{\text{out}}+1} = \boldsymbol{S}\boldsymbol{B}_{m_{\text{out}}+1}$, and $\boldsymbol{b}_{m_{\text{out}}+1}$ satisfies $\mathcal{C}$

$\beta = \|\boldsymbol{b}\|$, $\boldsymbol{b}_1 = \boldsymbol{b}/\beta$, $\boldsymbol{p}_1 = \boldsymbol{p}/\beta$
Let s be the number of rows of $\boldsymbol{S}$, and initialize $\boldsymbol{P}_1 = [\boldsymbol{p}_1]$
for $j = 1, 2, \ldots, m_{\max}$ **do**
  $\widetilde{\boldsymbol{b}}_{j+1} = \boldsymbol{A}\boldsymbol{b}_j$
  for $i = \max(1, j-t+1), \ldots, j$ **do**
    $h_{i,j} = \langle \widetilde{\boldsymbol{b}}_{j+1}, \boldsymbol{b}_i \rangle$, $\widetilde{\boldsymbol{b}}_{j+1} = \widetilde{\boldsymbol{b}}_{j+1} - \boldsymbol{b}_i h_{i,j}$
  end for
  $h_{j+1,j} = \|\widetilde{\boldsymbol{b}}_{j+1}\|$, $\boldsymbol{b}_{j+1} = \widetilde{\boldsymbol{b}}_{j+1}/h_{j+1,j}$
  $\boldsymbol{p}_{j+1} = \boldsymbol{S}\boldsymbol{b}_{j+1}$, $\boldsymbol{P}_{j+1} = [\boldsymbol{P}_j, \boldsymbol{p}_{j+1}]$
  if $j < m_{\max}$ **then**
    if $s < \eta(j+1)$ **then**
      $s \leftarrow s + s_0$
      Draw a new sketching matrix $\boldsymbol{S} \in \mathbb{C}^{s \times N}$
      $\boldsymbol{P}_{j+1} \leftarrow \boldsymbol{S}\boldsymbol{B}_{j+1}$
    end if
    if $\kappa(\boldsymbol{P}_{j+1}) > \tau_{\text{cond}}$ **then**
      Break
    end if
  end if
end for
Set $m_{\text{out}} = j$
Compute the rank-1 update from Section 2.3 for $\mathcal{C}$, using $\alpha_{m_{\text{out}}+1} = \|\boldsymbol{b}_{m_{\text{out}}+1} - \boldsymbol{B}_{m_{\text{out}}}\boldsymbol{c}_{m_{\text{out}}}\|$
$\boldsymbol{p}_{m_{\text{out}}+1} \leftarrow \big(\boldsymbol{p}_{m_{\text{out}}+1} - \boldsymbol{P}_{m_{\text{out}}}\boldsymbol{c}_{m_{\text{out}}}\big)/\alpha_{m_{\text{out}}+1}$
Replace $\boldsymbol{H}_{m_{\text{out}}}$, $\boldsymbol{b}_{m_{\text{out}}+1}$, and $h_{m_{\text{out}}+1,m_{\text{out}}}$ by their updated values

quadrature errors, so the analysis focuses on the Krylov and sketching components of the methods. The non-sketched adaptive variants can be related directly to existing restarted Arnoldi and harmonic Arnoldi methods through the Arnoldi-like framework developed above. The main new issue is the convergence of the restarted sketched harmonic Arnoldi method, for which the Petrov–Galerkin condition is imposed only after applying the sketching matrix.

Our analysis extends the GMRES-based approach of [11] to the sketched setting. This approach is particularly useful because it does not rely on CG-type arguments and therefore remains applicable in the presence of complex Ritz values. We focus on Stieltjes functions of the form

$$f(z) = \int_{-\infty}^{0} \frac{g(t)}{t - z}\, \mathrm{d}t, \tag{5.1}$$

Algorithm 4.4 Restarted sketched Krylov subspace methods for computing $f(\boldsymbol{A})\boldsymbol{b}$

Input: Matrix $\boldsymbol{A}$, vector $\boldsymbol{b}$, function f, subspace dimension m, sketching dimension s, choice of the s-Arnoldi or sHm-Arnoldi process, stopping tolerance τ, quadrature tolerance τ_{quad}, and maximum number of quadrature nodes $n_{\mathrm{quad,max}}$

Output: $\boldsymbol{f}^{[k]} \approx f(\boldsymbol{A})\boldsymbol{b}$

Draw a sketching matrix $\boldsymbol{S} \in \mathbb{C}^{s\times N}$ and set $\boldsymbol{p} = \boldsymbol{S}\boldsymbol{b}$
Compute $\beta^{[0]}$, $\boldsymbol{B}_{m+1}^{[0]}$, $\boldsymbol{P}_{m+1}^{[0]}$, and $\underline{\boldsymbol{H}}_m^{[0]}$ by calling `sArnoldi` or `sHmArnoldi` with inputs $(\boldsymbol{S}, \boldsymbol{A}, \boldsymbol{b}, \boldsymbol{p}, m)$, according to the selected process
$\boldsymbol{f}^{[0]} = \boldsymbol{B}_m^{[0]} f(\boldsymbol{H}_m^{[0]})\boldsymbol{e}_1\beta^{[0]}$
for $k = 1, 2, \ldots$ **do**
  Determine the error function $\mathrm{err}^{[k]}(z)$ using Corollary 3.4
  Compute $\beta^{[k]}$, $\boldsymbol{B}_{m+1}^{[k]}$, $\boldsymbol{P}_{m+1}^{[k]}$, and $\underline{\boldsymbol{H}}_m^{[k]}$ by calling `sArnoldi` or `sHmArnoldi` with inputs $(\boldsymbol{S}, \boldsymbol{A}, \boldsymbol{b}_{m+1}^{[k-1]}, \boldsymbol{p}_{m+1}^{[k-1]}, m)$, according to the selected process
  Compute $\boldsymbol{u}^{[k]} \approx \mathrm{err}^{[k]}(\boldsymbol{H}_m^{[k]})\boldsymbol{e}_1\beta^{[k]}$ using adaptive quadrature with tolerance τ_{quad} and maximum number of quadrature nodes $n_{\mathrm{quad,max}}$
  $\boldsymbol{g}^{[k]} = \boldsymbol{B}_m^{[k]}\boldsymbol{u}^{[k]}$
  $\boldsymbol{f}^{[k]} = \boldsymbol{f}^{[k-1]} + \boldsymbol{g}^{[k]}$
  if $\|\boldsymbol{g}^{[k]}\| < \tau \|\boldsymbol{f}^{[k]}\|$ **then**
    Break
  end if
end for

Algorithm 4.5 Restarted adaptive Krylov subspace methods for computing $f(\boldsymbol{A})\boldsymbol{b}$

Input: Matrix $\boldsymbol{A}$, vector $\boldsymbol{b}$, function f, truncation parameter t, condition number threshold τ_{cond}, maximum subspace dimension $m_{\max}$, sketching dimension increment s_0, sketching dimension control parameter η, orthogonality condition $\mathcal{C}$ from Section 2.3, stopping tolerance τ, quadrature tolerance τ_{quad}, and maximum number of quadrature nodes $n_{\mathrm{quad,max}}$

Output: $\boldsymbol{f}^{[k]} \approx f(\boldsymbol{A})\boldsymbol{b}$

Draw a sketching matrix $\boldsymbol{S} \in \mathbb{C}^{s_0\times N}$ and set $\boldsymbol{p} = \boldsymbol{S}\boldsymbol{b}$
$[\boldsymbol{S}, m_0, \beta^{[0]}, \boldsymbol{B}_{m_0+1}^{[0]}, \boldsymbol{P}_{m_0+1}^{[0]}, \underline{\boldsymbol{H}}_{m_0}^{[0]}]$
  $= \texttt{tArnoldi_adaptive}(\boldsymbol{S}, \boldsymbol{A}, \boldsymbol{b}, \boldsymbol{p}, m_{\max}, t, \tau_{\mathrm{cond}}, s_0, \eta, \mathcal{C})$
Set $m_{\max} = m_0$
$\boldsymbol{f}^{[0]} = \boldsymbol{B}_{m_0}^{[0]} f(\boldsymbol{H}_{m_0}^{[0]})\boldsymbol{e}_1\beta^{[0]}$
for $k = 1, 2, \ldots$ **do**
  Determine the error function $\mathrm{err}^{[k]}(z)$ using Corollary 3.4
  $[\boldsymbol{S}, m_k, \beta^{[k]}, \boldsymbol{B}_{m_k+1}^{[k]}, \boldsymbol{P}_{m_k+1}^{[k]}, \underline{\boldsymbol{H}}_{m_k}^{[k]}]$
    $= \texttt{tArnoldi_adaptive}(\boldsymbol{S}, \boldsymbol{A}, \boldsymbol{b}_{m_{k-1}+1}^{[k-1]}, \boldsymbol{p}_{m_{k-1}+1}^{[k-1]}, m_{\max}, t, \tau_{\mathrm{cond}}, s_0, \eta, \mathcal{C})$
  Compute $\boldsymbol{u}^{[k]} \approx \mathrm{err}^{[k]}(\boldsymbol{H}_{m_k}^{[k]})\boldsymbol{e}_1\beta^{[k]}$ using adaptive quadrature with tolerance τ_{quad} and maximum number of quadrature nodes $n_{\mathrm{quad,max}}$
  $\boldsymbol{g}^{[k]} = \boldsymbol{B}_{m_k}^{[k]}\boldsymbol{u}^{[k]}$
  $\boldsymbol{f}^{[k]} = \boldsymbol{f}^{[k-1]} + \boldsymbol{g}^{[k]}$
  if $\|\boldsymbol{g}^{[k]}\| < \tau \|\boldsymbol{f}^{[k]}\|$ **then**
    Break
  end if
end for

where g is a suitable nonpositive function on $(-\infty, 0]$.

5.1. Convergence of the adaptive restarted Krylov methods. We first consider the non-sketched adaptive variants of Algorithm 4.5. The t-Arnoldi basis may be nonorthonormal; the final rank-1 update enforces the selected Arnoldi or harmonic Arnoldi orthogonality condition, making these variants mathematically equivalent to the corresponding restarted methods in [11].

Assume that we have an Arnoldi-like decomposition (2.4), where $\boldsymbol{B}_m$ has full column rank but is not necessarily orthonormal. We further assume that $\boldsymbol{b}_{m+1} \perp \boldsymbol{A}\boldsymbol{B}_m$, which can be enforced by the rank-1 update discussed in Section 2.3. Under this assumption, Algorithm 4.5 is mathematically equivalent to the restarted HmArnoldi method analyzed in [11]. The corresponding HmArnoldi approximation to the solution of the shifted linear system (2.3) is $\boldsymbol{x}_m^{\mathrm{HmArn}}(t) = \boldsymbol{B}_m(t\boldsymbol{I}_m - \boldsymbol{H}_m)^{-1}\boldsymbol{e}_1\beta$. Its residual satisfies

$$\begin{aligned}
\boldsymbol{r}_m^{\mathrm{HmArn}}(t) &= \boldsymbol{b} - (t\boldsymbol{I} - \boldsymbol{A})\boldsymbol{x}_m^{\mathrm{HmArn}}(t) \\
&= \boldsymbol{b} - (t\boldsymbol{I} - \boldsymbol{A})\boldsymbol{B}_m(t\boldsymbol{I}_m - \boldsymbol{H}_m)^{-1}\boldsymbol{e}_1\beta \\
&= \boldsymbol{b} - [\boldsymbol{B}_m\ \boldsymbol{b}_{m+1}](t\underline{\boldsymbol{I}}_m - \underline{\boldsymbol{H}}_m)(t\boldsymbol{I}_m - \boldsymbol{H}_m)^{-1}\boldsymbol{e}_1\beta \\
&= \boldsymbol{b}_{m+1}h_{m+1,m}\boldsymbol{e}_m^\top(t\boldsymbol{I}_m - \boldsymbol{H}_m)^{-1}\boldsymbol{e}_1\beta \\
&=: \boldsymbol{b}_{m+1}\zeta^{\mathrm{HmArn}}(t).
\end{aligned}$$

Thus, the residual is collinear with $\boldsymbol{b}_{m+1}$ for all t.

For each t, let $p_{m-1,t}^{\mathrm{HmArn}} \in \mathcal{P}_{m-1}$ be the polynomial that interpolates $h_t(z) = (t-z)^{-1}$ at the harmonic Ritz values $\Lambda(\boldsymbol{H}_m)$. Then Lemma 3.2 gives

$$\boldsymbol{x}_m^{\mathrm{HmArn}}(t) = \boldsymbol{B}_m(t\boldsymbol{I}_m - \boldsymbol{H}_m)^{-1}\boldsymbol{e}_1\beta = p_{m-1,t}^{\mathrm{HmArn}}(\boldsymbol{A})\boldsymbol{b}.$$

If f is a Stieltjes function of the form (5.1), then the HmArnoldi approximation (2.8) can be written as

$$\boldsymbol{f}_m^{\mathrm{HmArn}} = \int_{-\infty}^{0} g(t)\boldsymbol{x}_m^{\mathrm{HmArn}}(t)\,\mathrm{d}t$$

. Up to the sign convention in the Stieltjes representation, this is precisely the class of HmArnoldi approximants analyzed in [11]. Consequently, the convergence results of [11] carry over to Algorithm 4.5. The following theorem summarizes this observation.

THEOREM 5.1 (Convergence of the adaptive restarted Arnoldi-like method). *Let f be a Stieltjes function, and consider Algorithm 4.5, where the basis is generated by the truncated Arnoldi process in Algorithm 4.3.*

If the rank-1 update enforces $\boldsymbol{b}_{m+1} \perp \boldsymbol{B}_m$, then the algorithm is mathematically equivalent to the restarted Arnoldi method for computing $f(\boldsymbol{A})\boldsymbol{b}$. Hence, if $\boldsymbol{A}$ is Hermitian positive definite, the method converges for every restart length $m \geq 1$.

If the rank-1 update enforces $\boldsymbol{b}_{m+1} \perp \boldsymbol{A}\boldsymbol{B}_m$, then the algorithm is mathematically equivalent to the restarted HmArnoldi method for computing $f(\boldsymbol{A})\boldsymbol{b}$. Hence, if $\boldsymbol{A}$ is positive real, the method converges for every restart length $m \geq 1$.

Proof. The equivalence to the restarted Arnoldi and restarted HmArnoldi methods follows from the Arnoldi-like framework and the rank-1 update described above. The convergence of the restarted Arnoldi method for Hermitian positive definite matrices is established in [11, Theorem 4.3], while the convergence of the restarted

HmArnoldi method for positive real matrices is proved in [11, Theorem 6.5]. Since both convergence results hold for every restart length $m \geq 1$, they remain valid when the restart length varies between restart cycles. ∎

5.2. Sketched harmonic Arnoldi with a nonorthogonal basis. We now turn to the sketched setting, where the Petrov–Galerkin condition is imposed only after applying the sketching matrix. Unlike the non-sketched variants discussed above, the restarted sHmArnoldi method cannot be reduced directly to the HmArnoldi method analyzed in [11]. We therefore establish the residual relations needed to extend the GMRES-based convergence analysis to the sketched setting.

Suppose that, in the Arnoldi-like decomposition (2.4), the matrix $\boldsymbol{B}_m$ has full column rank but is not necessarily orthonormal. Assume further that $\boldsymbol{b}_{m+1} \perp_{\boldsymbol{S}} \boldsymbol{A}\boldsymbol{B}_m$. The sHmArnoldi approximation to the solution of the shifted linear system (2.3) is $\boldsymbol{x}_m^{\mathrm{sHmArn}}(t) = \boldsymbol{B}_m(t\boldsymbol{I}_m - \boldsymbol{H}_m)^{-1}\boldsymbol{e}_1\beta$. As in Section 5.1, its residual satisfies

$$\boldsymbol{r}_m^{\mathrm{sHmArn}}(t) = \boldsymbol{b} - (t\boldsymbol{I} - \boldsymbol{A})\boldsymbol{x}_m^{\mathrm{sHmArn}}(t) = \boldsymbol{b}_{m+1}\zeta^{\mathrm{sHmArn}}(t). \tag{5.2}$$

Thus, the residual is collinear with $\boldsymbol{b}_{m+1}$ for all t.

Similarly, for each t, there exists a polynomial $p_{m-1,t}^{\mathrm{sHmArn}} \in \mathcal{P}_{m-1}$ such that

$$\boldsymbol{x}_m^{\mathrm{sHmArn}}(t) = p_{m-1,t}^{\mathrm{sHmArn}}(\boldsymbol{A})\boldsymbol{b},$$

where $p_{m-1,t}^{\mathrm{sHmArn}}$ interpolates $h_t(z) = (t-z)^{-1}$ at the sketched harmonic Ritz values $\Lambda(\boldsymbol{H}_m)$. Therefore, the residual can be written as

$$\boldsymbol{r}_m^{\mathrm{sHmArn}}(t) = q_{m,t}^{\mathrm{sHmArn}}(\boldsymbol{A})\boldsymbol{b}, \qquad q_{m,t}^{\mathrm{sHmArn}}(z) = 1 - (t-z)p_{m-1,t}^{\mathrm{sHmArn}}(z).$$

The following lemma is the sketched analogue of [11, Lemma 6.2(iv)]. We include a direct proof for completeness.

LEMMA 5.2 (A formula for the residual). *Let $\Lambda(\boldsymbol{H}_m) = \{\theta_1, \ldots, \theta_m\}$, and assume that $\theta_j \neq 0$ for $j = 1, \ldots, m$. Then the residuals defined in* (5.2) *satisfy*

$$\boldsymbol{r}_m^{\mathrm{sHmArn}}(t) = \frac{1}{(1 - t/\theta_1)\cdots(1 - t/\theta_m)}\boldsymbol{r}_m^{\mathrm{sHmArn}}(0) =: \boldsymbol{r}_m^{\mathrm{sHmArn}}(0)\eta_m(t).$$

Proof. Let $\phi_m(z) = (z - \theta_1)\cdots(z - \theta_m)$. It is straightforward to verify that $p_{m-1,t}^{\mathrm{sHmArn}}(z) = \big(1 - \phi_m(z)/\phi_m(t)\big)/(t - z)$ is the polynomial interpolating $h_t(z) = (t-z)^{-1}$ at $\Lambda(\boldsymbol{H}_m)$. Direct substitution gives $q_{m,t}^{\mathrm{sHmArn}}(z) = 1 - (t-z)p_{m-1,t}^{\mathrm{sHmArn}}(z) = \phi_m(z)/\phi_m(t)$. Therefore,

$$\boldsymbol{r}_m^{\mathrm{sHmArn}}(t) = q_{m,t}^{\mathrm{sHmArn}}(\boldsymbol{A})\boldsymbol{b} = \frac{\phi_m(\boldsymbol{A})}{\phi_m(t)}\boldsymbol{b} = \frac{\phi_m(0)}{\phi_m(t)}\frac{\phi_m(\boldsymbol{A})}{\phi_m(0)}\boldsymbol{b} = \frac{\phi_m(0)}{\phi_m(t)}\boldsymbol{r}_m^{\mathrm{sHmArn}}(0).$$

Since $\phi_m(t)/\phi_m(0) = (1 - t/\theta_1)\cdots(1 - t/\theta_m)$, the stated formula follows. ∎

Suppose that $\boldsymbol{A}$ is positive real. If the sketched harmonic Ritz values $\Lambda(\boldsymbol{H}_m)$ lie in the open right half-plane $\mathbb{C}_+$, the modulus of the factor $\phi_m(0)/\phi_m(t)$ decreases as t moves from 0 toward $-\infty$. Moreover, $\boldsymbol{r}_m^{\mathrm{sHmArn}}(0)$ is precisely the residual of the sGMRES approximant $\boldsymbol{x}_m^{\mathrm{sHmArn}}(0)$. The following lemma provides a bound for $\|\boldsymbol{r}_m^{\mathrm{sHmArn}}(0)\|$.

LEMMA 5.3 (Bound on the residual). *Let $\boldsymbol{A} \in \mathbb{C}^{N\times N}$ be positive real, and let $\boldsymbol{b} \in \mathbb{C}^N$. Assume that the sketching matrix $\boldsymbol{S}$ satisfies* (2.9) *for some $\varepsilon \in [0, 1)$.*

Let $\boldsymbol{x}_m^{\mathrm{sHmArn}}(t)$ be the sHmArnoldi approximation to the solution of the shifted linear system (2.3)*, and let $\boldsymbol{r}_m^{\mathrm{sHmArn}}(t)$ be the corresponding residual defined in* (5.2)*. Then*

$$\|\boldsymbol{r}_m^{\mathrm{sHmArn}}(0)\| \le C_\varepsilon \alpha_m \|\boldsymbol{b}\|,$$

where $C_\varepsilon = \sqrt{(1+\varepsilon)/(1-\varepsilon)}$ and $\alpha_m = (1-\delta\delta')^{m/2} \in [0,1)$. The constants δ and δ' are defined by

$$\begin{aligned} \delta &= \min\bigl\{|\boldsymbol{v}^*\boldsymbol{A}\boldsymbol{v}| : \boldsymbol{v}\in\mathbb{C}^N,\ \|\boldsymbol{v}\|=1\bigr\}, \\ \delta' &= \min\bigl\{|\boldsymbol{v}^*\boldsymbol{A}^{-1}\boldsymbol{v}| : \boldsymbol{v}\in\mathbb{C}^N,\ \|\boldsymbol{v}\|=1\bigr\}. \end{aligned} \tag{5.3}$$

Proof. Let $\boldsymbol{x}_m^{\mathrm{HmArn}}(t)$ be the HmArnoldi approximation to the solution of the shifted linear system (2.3), and let $\boldsymbol{r}_m^{\mathrm{HmArn}}(t)$ be the corresponding residual. By [11, Lemma 6.3], $\|\boldsymbol{r}_m^{\mathrm{HmArn}}(0)\| \le \alpha_m\|\boldsymbol{b}\|$. Since the subspace embedding property holds for the relevant residual vectors, by the sketched residual minimization property of sGMRES, we have

$$\begin{aligned} \|\boldsymbol{r}_m^{\mathrm{sHmArn}}(0)\| &\le \frac{1}{\sqrt{1-\varepsilon}}\|\boldsymbol{S}\boldsymbol{r}_m^{\mathrm{sHmArn}}(0)\| \le \frac{1}{\sqrt{1-\varepsilon}}\|\boldsymbol{S}\boldsymbol{r}_m^{\mathrm{HmArn}}(0)\| \\ &\le \sqrt{\frac{1+\varepsilon}{1-\varepsilon}}\|\boldsymbol{r}_m^{\mathrm{HmArn}}(0)\|. \end{aligned}$$

Combining the above estimate with $\|\boldsymbol{r}_m^{\mathrm{HmArn}}(0)\| \le \alpha_m\|\boldsymbol{b}\|$ completes the proof. ∎

For a Stieltjes function f of the form (5.1), the sHmArnoldi approximation to $f(\boldsymbol{A})\boldsymbol{b}$ can be written as

$$\boldsymbol{f}_m^{\mathrm{sHmArn}} = \int_{-\infty}^0 g(t)\boldsymbol{x}_m^{\mathrm{sHmArn}}(t)\,\mathrm{d}t.$$

Consequently,

$$\begin{aligned} f(\boldsymbol{A})\boldsymbol{b} - \boldsymbol{f}_m^{\mathrm{sHmArn}} &= \int_{-\infty}^0 g(t)\bigl(\boldsymbol{x}(t) - \boldsymbol{x}_m^{\mathrm{sHmArn}}(t)\bigr)\,\mathrm{d}t \\ &= \int_{-\infty}^0 g(t)(t\boldsymbol{I}-\boldsymbol{A})^{-1}\bigl(\boldsymbol{b} - (t\boldsymbol{I}-\boldsymbol{A})\boldsymbol{x}_m^{\mathrm{sHmArn}}(t)\bigr)\,\mathrm{d}t \\ &= \int_{-\infty}^0 g(t)(t\boldsymbol{I}-\boldsymbol{A})^{-1}\boldsymbol{r}_m^{\mathrm{sHmArn}}(t)\,\mathrm{d}t \\ &= \int_{-\infty}^0 g(t)(t\boldsymbol{I}-\boldsymbol{A})^{-1}\eta_m(t)\boldsymbol{r}_m^{\mathrm{sHmArn}}(0)\,\mathrm{d}t \\ &= \mathrm{err}(\boldsymbol{A})\boldsymbol{r}_m^{\mathrm{sHmArn}}(0), \end{aligned}$$

where $\mathrm{err}(z) = \int_{-\infty}^0 g(t)\eta_m(t)(t-z)^{-1}\,\mathrm{d}t$. Combining the above discussion, we obtain the convergence of the restarted sHmArnoldi method for computing $f(\boldsymbol{A})\boldsymbol{b}$ when $\boldsymbol{A}$ is positive real and f is a Stieltjes function.

THEOREM 5.4 (Convergence of the restarted sHmArnoldi method). *Let $\boldsymbol{A} \in \mathbb{C}^{N\times N}$ be positive real, let $\boldsymbol{b}\in\mathbb{C}^N$, and let f be a Stieltjes function of the form* (5.1)*. Fix an integer $k \ge 0$. Assume that the sketching matrix $\boldsymbol{S}$, reused in every restart cycle, satisfies the norm inequalities in* (2.9) *with a common $\varepsilon \in [0,1)$ for every*

$0 \le \ell \le k$ and every $\boldsymbol{v} \in \mathcal{K}_{m+1}(\boldsymbol{A}, \boldsymbol{b}_1^{[\ell]})$. Here, $\boldsymbol{b}_1^{[\ell]}$ is the first basis vector in the ℓ-th cycle. Assume also that the sketched harmonic Ritz values satisfy $\theta_j^{[\ell]} \in \mathbb{C}_+$ for $0 \le \ell \le k$ and $1 \le j \le m$. Then the restarted sHmArnoldi approximation computed by Algorithm 4.4 satisfies

$$\big\| f(\boldsymbol{A})\boldsymbol{b} - \boldsymbol{f}^{\mathrm{sHmArn},[k]} \big\|_{\boldsymbol{A}^*\boldsymbol{A}} \le C\|\boldsymbol{b}\| \big(C_\varepsilon \alpha_m\big)^{k+1}, \tag{5.4}$$

where $C = \sqrt{\nu_{\max}} f(\rho\nu_{\max})$, and α_m and C_ε are defined in Lemma 5.3. Here,

$$\begin{aligned} \nu_{\max} &= \max\big\{ \boldsymbol{v}^*\boldsymbol{A}^*\boldsymbol{A}\boldsymbol{v} : \boldsymbol{v} \in \mathbb{C}^N,\ \|\boldsymbol{v}\| = 1 \big\} = \|\boldsymbol{A}\|^2, \\ \rho &= \min\big\{ \Re\big(\boldsymbol{v}^*\boldsymbol{A}^{-1}\boldsymbol{v}\big) : \boldsymbol{v} \in \mathbb{C}^N,\ \|\boldsymbol{v}\| = 1 \big\}. \end{aligned} \tag{5.5}$$

Since $\boldsymbol{A}$ is positive real, we have $\rho > 0$. Thus, $\rho\nu_{\max} > 0$, and the representation (5.1) *with g nonpositive implies that $f(\rho\nu_{\max}) > 0$.*

Proof. By Lemma 5.2, the residual in the ℓ-th cycle satisfies $\boldsymbol{r}_m^{\mathrm{sHmArn},[\ell]}(t) = \boldsymbol{r}_m^{\mathrm{sHmArn},[\ell]}(0)\eta_m^{[\ell]}(t)$. Following the argument in Corollary 3.4, the remaining error after k restart cycles can be written as

$$f(\boldsymbol{A})\boldsymbol{b} - \boldsymbol{f}^{\mathrm{sHmArn},[k]} = \left(\int_{-\infty}^{0} g(t) \prod_{\ell=0}^{k} \eta_m^{[\ell]}(t) (t\boldsymbol{I} - \boldsymbol{A})^{-1} \,\mathrm{d}t \right) \boldsymbol{r}_m^{\mathrm{sHmArn},[k]}(0).$$

Since $\Re(\theta_j^{[\ell]}) > 0$, we have $\Re(-t/\theta_j^{[\ell]}) \ge 0$ for all $t \le 0$. Hence, $|1 - t/\theta_j^{[\ell]}| \ge 1$, and

$$|\eta_m^{[\ell]}(t)| = \prod_{j=1}^{m} \frac{1}{|1 - t/\theta_j^{[\ell]}|} \le 1, \qquad 0 \le \ell \le k, \quad t \le 0.$$

Following the proof of [11, Theorem 6.5] and using $|\eta_m^{[\ell]}(t)| \le 1$, we obtain

$$\begin{aligned} \big\| f(\boldsymbol{A})\boldsymbol{b} - \boldsymbol{f}^{\mathrm{sHmArn},[k]} \big\|_{\boldsymbol{A}^*\boldsymbol{A}} &\le \|\boldsymbol{r}_m^{\mathrm{sHmArn},[k]}(0)\| \int_{-\infty}^{0} g(t) \frac{\sqrt{\nu_{\max}}}{t - \rho\nu_{\max}} \,\mathrm{d}t \\ &= \|\boldsymbol{r}_m^{\mathrm{sHmArn},[k]}(0)\| \sqrt{\nu_{\max}} f(\rho\nu_{\max}). \end{aligned}$$

Here, we used the fact that g is nonpositive in the Stieltjes representation (5.1). Finally, Lemma 5.3 gives $\|\boldsymbol{r}_m^{\mathrm{sHmArn},[k]}(0)\| \le C_\varepsilon \alpha_m \|\boldsymbol{r}_m^{\mathrm{sHmArn},[k-1]}(0)\|$. Applying this inequality recursively and defining $C = \sqrt{\nu_{\max}} f(\rho\nu_{\max})$ completes the proof. ∎

As explained in Section 5.1, the non-sketched adaptive variants inherit convergence for Stieltjes functions and every restart length $m \ge 1$, with Hermitian positive definite matrices for Arnoldi and positive real matrices for HmArnoldi [11, Theorems 4.3 and 6.5]. In the latter setting, Theorem 5.4 adds embedding and sketched harmonic Ritz-value assumptions and replaces the contraction factor α_m by $C_\varepsilon \alpha_m$.

The factor C_ε compares sketched and Euclidean residual norms and depends on embedding distortion, independently of basis orthonormality. If $\varepsilon = 0$ on the relevant subspace in every cycle, sketching preserves inner products, recovering Euclidean harmonic orthogonality, the HmArnoldi approximants, and their bound with $C_\varepsilon = 1$.

In exact arithmetic and neglecting quadrature errors, geometric convergence follows if the embedding and Ritz-value assumptions hold in every cycle with a common ε satisfying $C_\varepsilon \alpha_m < 1$. Failure of this sufficient inequality does not imply divergence, and the bound is not intended to predict the observed rate.

5.3. Sketched harmonic Ritz values. The convergence result in Theorem 5.4 assumes that all sketched harmonic Ritz values lie in the open right half-plane. This subsection provides a sufficient condition under which this assumption holds. The argument relies only on the orthogonality condition produced by the sHm-Arnoldi process, namely, $\boldsymbol{b}_{m+1} \perp_{\boldsymbol{S}} \boldsymbol{A}\boldsymbol{B}_m$.

Denote $\boldsymbol{Z}_m = \boldsymbol{A}\boldsymbol{B}_m$. Since $\boldsymbol{b}_{m+1} \perp_{\boldsymbol{S}} \boldsymbol{Z}_m$, multiplying the Arnoldi-like decomposition (2.4) from the left by $(\boldsymbol{S}\boldsymbol{Z}_m)^*\boldsymbol{S}$ gives

$$(\boldsymbol{S}\boldsymbol{Z}_m)^*(\boldsymbol{S}\boldsymbol{Z}_m) = (\boldsymbol{S}\boldsymbol{Z}_m)^*(\boldsymbol{S}\boldsymbol{B}_m)\boldsymbol{H}_m.$$

Therefore, $\boldsymbol{H}_m = \boldsymbol{G}_m^{-1}\boldsymbol{F}_m$, where

$$\boldsymbol{G}_m = (\boldsymbol{S}\boldsymbol{Z}_m)^*(\boldsymbol{S}\boldsymbol{B}_m) \quad \text{and} \quad \boldsymbol{F}_m = (\boldsymbol{S}\boldsymbol{Z}_m)^*(\boldsymbol{S}\boldsymbol{Z}_m),$$

provided that $\boldsymbol{G}_m$ is nonsingular. Thus, the sketched harmonic Ritz values are the generalized eigenvalues of the matrix pencil $(\boldsymbol{F}_m, \boldsymbol{G}_m)$.

Let $\theta \in \Lambda(\boldsymbol{H}_m)$, and let $\boldsymbol{y} \in \mathbb{C}^m$ be a corresponding nonzero eigenvector. Set $\boldsymbol{x} = \boldsymbol{Z}_m\boldsymbol{y}$. Then $\boldsymbol{x} \neq \boldsymbol{0}$ and $\boldsymbol{F}_m\boldsymbol{y} = \boldsymbol{G}_m\boldsymbol{y}\theta$. This gives

$$\theta = \frac{\boldsymbol{y}^*\boldsymbol{F}_m\boldsymbol{y}}{\boldsymbol{y}^*\boldsymbol{G}_m\boldsymbol{y}} = \frac{\|\boldsymbol{S}\boldsymbol{x}\|^2}{(\boldsymbol{S}\boldsymbol{x})^*(\boldsymbol{S}\boldsymbol{A}^{-1}\boldsymbol{x})}.$$

Equivalently,

$$\frac{1}{\theta} = \frac{(\boldsymbol{S}\boldsymbol{x})^*(\boldsymbol{S}\boldsymbol{A}^{-1}\boldsymbol{x})}{\|\boldsymbol{S}\boldsymbol{x}\|^2}. \tag{5.6}$$

Taking $\boldsymbol{u} = \boldsymbol{A}^{-1}\boldsymbol{x} = \boldsymbol{B}_m\boldsymbol{y}$ and $\boldsymbol{v} = \boldsymbol{x} = \boldsymbol{A}\boldsymbol{B}_m\boldsymbol{y}$ in the inner-product form of the subspace embedding property (2.10), we obtain

$$\left|(\boldsymbol{S}\boldsymbol{x})^*(\boldsymbol{S}\boldsymbol{A}^{-1}\boldsymbol{x}) - \boldsymbol{x}^*\boldsymbol{A}^{-1}\boldsymbol{x}\right| \leq \varepsilon\|\boldsymbol{A}^{-1}\boldsymbol{x}\|\|\boldsymbol{x}\| \leq \varepsilon\|\boldsymbol{A}^{-1}\|\|\boldsymbol{x}\|^2.$$

Combining this estimate with the subspace embedding property (2.9) and the identity (5.6), we conclude that the reciprocal of every sketched harmonic Ritz value satisfies

$$\frac{1}{\theta} \in \left\{ \frac{z_1 + z_2}{\tau} : z_1 \in \mathcal{W}(\boldsymbol{A}^{-1}),\ |z_2| \leq \varepsilon\|\boldsymbol{A}^{-1}\|,\ 1-\varepsilon \leq \tau \leq 1+\varepsilon \right\}.$$

Since $\boldsymbol{A}$ is positive real, $\boldsymbol{A}^{-1}$ is also positive real. Let ρ be defined as in (5.5). If $\varepsilon\|\boldsymbol{A}^{-1}\| < \rho$, then, for any sketched harmonic Ritz value θ, the preceding inclusion gives

$$\Re\left(\frac{1}{\theta}\right) \geq \frac{\rho - \varepsilon\|\boldsymbol{A}^{-1}\|}{1+\varepsilon} > 0.$$

Since $\Re(1/\theta) > 0$ is equivalent to $\Re(\theta) > 0$, all sketched harmonic Ritz values lie in the open right half-plane. Moreover, under the condition $\varepsilon\|\boldsymbol{A}^{-1}\| < \rho$, for every nonzero $\boldsymbol{y} \in \mathbb{C}^m$, with $\boldsymbol{x} = \boldsymbol{A}\boldsymbol{B}_m\boldsymbol{y}$, we have

$$\Re\big(\boldsymbol{y}^*\boldsymbol{G}_m\boldsymbol{y}\big) = \Re\big((\boldsymbol{S}\boldsymbol{x})^*(\boldsymbol{S}\boldsymbol{A}^{-1}\boldsymbol{x})\big) \geq \big(\rho - \varepsilon\|\boldsymbol{A}^{-1}\|\big)\|\boldsymbol{x}\|^2 > 0.$$

Hence, $\boldsymbol{G}_m$ is nonsingular, and the condition $\varepsilon\|\boldsymbol{A}^{-1}\| < \rho$ is sufficient for the sketched harmonic Ritz values generated by the sHm-Arnoldi process to lie in $\mathbb{C}_+$, as required in Theorem 5.4.

Combining this condition with $C_\varepsilon \alpha_m < 1$ gives the joint sufficient restriction

$$0 \le \varepsilon < \min\left\{\frac{1-\alpha_m^2}{1+\alpha_m^2}, \frac{\rho}{\|\boldsymbol{A}^{-1}\|}\right\},$$

provided that (2.9) holds in every restart cycle with this common ε. The required distortion depends on $\boldsymbol{A}$ and, through α_m, on the restart length m. Choosing s balances this problem-dependent accuracy requirement against the cost of sketching.

6. Numerical experiments. In this section, we present numerical experiments to demonstrate the stability and efficiency of the proposed sketch-and-restart algorithms. All algorithms are implemented in MATLAB R2023b, and all experiments are performed on a server equipped with two Intel Xeon Gold 6226R CPUs running at 2.90 GHz and 1 TB of RAM. The code for reproducing the numerical experiments is available at https://github.com/JingyuLiuMath/funm_quad_sketch_and_restart.

The two matrix functions used in the experiments are

$$f_1(z) = z^{-1/2} = \frac{-1}{\pi}\int_{-\infty}^{0} \frac{(-t)^{-1/2}}{t-z}\,\mathrm{d}t$$

and

$$f_2(z) = \frac{\log(1+z)}{z} = \int_{-\infty}^{-1} \frac{t^{-1}}{t-z}\,\mathrm{d}t.$$

We use the sparse sign matrices described in Section 2.2, with $\zeta = 8$ following [17]. For this fixed sparsity, applying $\boldsymbol{S}$ to a vector costs $\mathscr{O}(N)$ operations.

The variants of Algorithm 4.4 based on the s-Arnoldi and sHm-Arnoldi processes are denoted by sArn and sHmArn, respectively; both use a fixed Krylov subspace dimension m and sketching dimension $s = 4m$. For comparison, we use the quadrature-based restarted Arnoldi method of [12] and its harmonic counterpart, denoted by Arn and HmArn. These use the standard Arnoldi process [21] and a deterministic variant of Algorithm 4.2, respectively.

For Algorithm 4.5, we set $t \in \{2, 1, 0\}$, $\tau_{\mathrm{cond}} = 10^6$, $s_0 = 30$, and $\eta = 4$. Each new vector is orthogonalized against at most the t most recent basis vectors; $t = 0$ omits these steps but retains normalization and the final rank-1 update. This update enforces $\boldsymbol{b}_{m+1} \perp \boldsymbol{B}_m$ or $\boldsymbol{b}_{m+1} \perp \boldsymbol{A}\boldsymbol{B}_m$, giving the methods adaArn-t and adaHmArn-t, respectively; see Section 2.3.

All methods use quadrature tolerance $\tau_{\mathrm{quad}} = 10^{-9}$, stopping tolerance $\tau = 10^{-8}$, and at most $n_{\mathrm{quad,max}} = 1024$ quadrature nodes. The stopping test compares the norm of the correction with τ times that of the current approximation.

We report the relative error $\|\boldsymbol{f}^{[k]} - \boldsymbol{f}_{\mathrm{ref}}\|_2 / \|\boldsymbol{f}_{\mathrm{ref}}\|_2$, where the reference vector $\boldsymbol{f}_{\mathrm{ref}}$ is supplied with the MATLAB code accompanying [15] for Section 6.1, computed using `sqrtm` applied to the full matrix for Section 6.2, or obtained from a known spectral decomposition of $\boldsymbol{A}$ for Section 6.3. The quantities n_{iter}, n_{matvec}, and m_{act} denote the number of restart cycles, the total number of matrix-vector products, and the average actual Krylov subspace dimension for each cycle, respectively.

To improve convergence, we also consider thick restarting, which retains a small approximate invariant subspace across cycles to preserve spectral information and can

improve convergence through approximate deflation [10]. For the adaptive Arnoldi methods, we retain lifted Schur vectors associated with Ritz values having the smallest absolute real parts. The target retention dimension is 5, increased when necessary to preserve a conjugate pair. Our current implementation of the harmonic Arnoldi processes does not support thick restarting, so this comparison is restricted to the Arnoldi variants.

We run each configuration three times with different random seeds and select the run with the median total runtime. All results reported in the tables and figures are taken from this selected run. All timings are reported in seconds.

6.1. Lattice quantum chromodynamics example. This example arises from lattice quantum chromodynamics (QCD), where one needs to compute $\operatorname{sign}(\boldsymbol{Q})$ on a vector for a large sparse non-Hermitian matrix $\boldsymbol{Q} \in \mathbb{C}^{N\times N}$. Since forming $\operatorname{sign}(\boldsymbol{Q})$ explicitly is infeasible, its action on a vector is computed as

$$\operatorname{sign}(\boldsymbol{Q})\boldsymbol{v} = (\boldsymbol{Q}^2)^{-1/2}\boldsymbol{Q}\boldsymbol{v} = f_1(\boldsymbol{A})\boldsymbol{b},$$

where $\boldsymbol{A} = \boldsymbol{Q}^2$ can be accessed only through matrix-vector products. In our experiment, $N = 49152$, $\boldsymbol{v} = \boldsymbol{e}_1/\|\boldsymbol{Q}\boldsymbol{e}_1\|$, so that $\boldsymbol{b} = \boldsymbol{Q}\boldsymbol{v} = \boldsymbol{Q}\boldsymbol{e}_1/\|\boldsymbol{Q}\boldsymbol{e}_1\|$. We choose $m = 150$ for fixed-dimension methods and $m_{\max} = 150$ for adaptive methods.

Table 2: Relative error, runtime in seconds, number of restart cycles n_{iter}, number of matrix-vector products n_{matvec}, and average actual Krylov subspace dimension m_{act} per restart cycle for the QCD example. The column "speedup" reports the percentage runtime reduction $(t_{\text{base}} - t_{\text{curr}})/t_{\text{base}} \times 100\%$, where the baseline is Arn (left group) or HmArn (right group). For fixed-dimension methods, $m = 150$ and $s = 4m = 600$. For adaptive methods, $m_{\max} = 150$ and $\eta = 4$. The row corresponding to the fastest method in each group is highlighted in bold.

method	error	time	n_{iter}	n_{matvec}	m_{act}	speedup	method	error	time	n_{iter}	n_{matvec}	m_{act}	speedup
Arn	1.5e-12	18.4	4	600	150.0	0.0%	HmArn	5.6e-12	17.7	4	600	150.0	0.0%
sArn	4.4e-12	16.2	4	600	150.0	12.2%	sHmArn	7.0e-12	16.7	4	600	150.0	5.2%
adaArn-2	**2.5e-10**	**12.9**	**5**	**514**	**102.8**	**29.8%**	adaHmArn-2	2.2e-11	15.4	6	609	101.5	13.1%
adaArn-1	4.0e-8	15.2	46	788	17.1	17.5%	**adaHmArn-1**	**4.6e-8**	**9.8**	**26**	**483**	**18.6**	**44.5%**
adaArn-0	1.4e-7	25.8	141	1269	9.0	−40.0%	adaHmArn-0	1.5e-7	27.2	140	1260	9.0	−54.0%

Table 2 and Figure 1 show that adaArn-2 and adaHmArn-1 are the fastest methods in the Arnoldi and harmonic Arnoldi groups, respectively. Their runtimes are 12.9 and 9.8 seconds, corresponding to runtime reductions of 29.8% and 44.5% relative to Arn and HmArn, respectively. They also require the fewest matrix-vector products in their respective groups: 514 and 483. Their final relative errors are 2.5×10^{-10} and 4.6×10^{-8}, respectively.

The methods sArn and sHmArn closely follow the convergence of Arn and HmArn, respectively. All four fixed-dimension methods require 4 restart cycles and 600 matrix-vector products. Each cycle generates 150 basis vectors before the stopping test is applied. Consequently, these methods require relatively few cycles but attain final relative errors of order 10^{-12}, far below the stopping tolerance 10^{-8}.

Figure 2 shows that basis generation accounts for most of the total runtime. The harmonic Arnoldi variants exhibit a similar runtime breakdown, so only the Arnoldi variants are shown. In the runtime breakdown, "Coefficients" refers to the computation of the inner products between Krylov vectors, and "Subtraction" refers to the vector-vector updates incurred during the basis construction. For sArn, applying

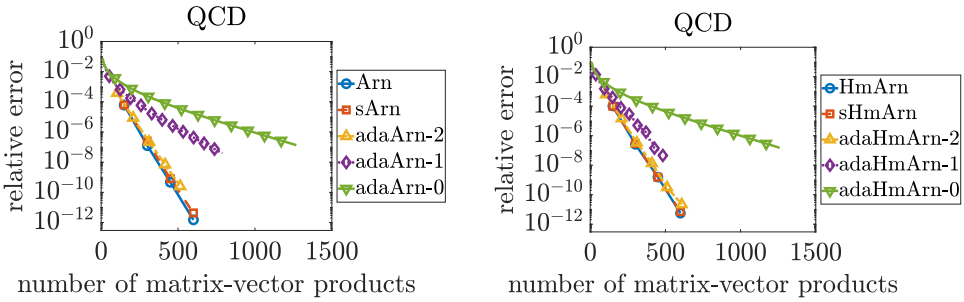


Fig. 1: Relative error versus the number of matrix-vector products for the QCD example. For fixed-dimension methods, $m = 150$ and $s = 4m = 600$. For adaptive methods, $m_{\max} = 150$ and $\eta = 4$.

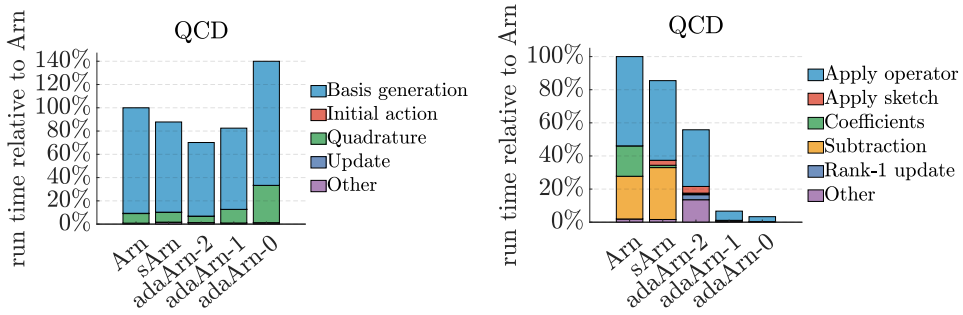


Fig. 2: Runtime breakdown for the QCD example. The left panel shows the total runtime, normalized by that of Arn. The right panel shows the average runtime per Arnoldi call, normalized by the corresponding average for Arn. For fixed-dimension methods, $m = 150$ and $s = 4m = 600$. For adaptive methods, $m_{\max} = 150$ and $\eta = 4$.

the sketch and computing the orthogonalization coefficients have nearly negligible costs. However, matrix-vector products with $\boldsymbol{A}$ still account for a large fraction of the basis-generation time, and updating the full-dimensional basis vectors remains necessary. These costs limit the overall savings from sketching: sArn and sHmArn reduce the runtime by only 12.2% and 5.2% relative to Arn and HmArn, respectively.

Decreasing t reduces the average actual Krylov dimension m_{act} and increases the number of restart cycles n_{iter}, but does not necessarily increase the runtime. The smaller subspaces make basis generation cheaper per cycle, which can compensate for the additional cycles. For example, reducing t from 2 to 1 in the harmonic group increases the number of cycles from 6 to 26, but reduces the runtime from 15.4 to 9.8 seconds. Further reducing t to 0 gives an average dimension of 9, but approximately 140 cycles are needed, making both adaptive methods slower than their respective baselines. Thus, the runtime depends on the balance between the cost per cycle and the number of cycles. The stopping test monitors the relative correction, so the final relative error need not be below 10^{-8}.

6.2. Fractional graph Laplacian example. In this example, $\boldsymbol{W}$ is the non-symmetric adjacency matrix of size $N = 62586$ corresponding to the directed graph `p2p_Gnutella31` from the SuiteSparse Matrix Collection [7]. We define the in-degree graph Laplacian by $\boldsymbol{A} = \boldsymbol{D} - \boldsymbol{W}$ and aim to compute $\boldsymbol{A}^{1/2}\boldsymbol{v}$ for a given vector $\boldsymbol{v} \in \mathbb{C}^N$. The matrix $\boldsymbol{A}$ is singular, and its spectrum lies in the closed right half-plane. We interpret the inverse square root as acting on the invariant subspace associated with the nonzero spectrum and compute

$$\boldsymbol{A}^{1/2}\boldsymbol{v} = \boldsymbol{A}^{-1/2}\boldsymbol{A}\boldsymbol{v} = f_1(\boldsymbol{A})\boldsymbol{b},$$

where $\boldsymbol{b} = \boldsymbol{A}\boldsymbol{v}$. Since $\boldsymbol{A}\boldsymbol{v} \in \mathcal{R}(\boldsymbol{A})$, the Krylov subspace methods considered here can be applied to this inverse-square-root action. In our experiments, both the real and imaginary parts of $\boldsymbol{v}$ are drawn from the standard normal distribution. We set $m = 150$ in this example.

Table 3: Relative error, runtime in seconds, number of restart cycles n_{iter}, number of matrix-vector products n_{matvec}, and average actual Krylov subspace dimension m_{act} per restart cycle for the graph Laplacian example. The column "speedup" reports the percentage runtime reduction $(t_{\text{base}} - t_{\text{curr}})/t_{\text{base}} \times 100\%$, where the baseline is Arn (left group) or HmArn (right group). For fixed-dimension methods, $m = 150$ and $s = 4m = 600$. For adaptive methods, $m_{\max} = 150$ and $\eta = 4$. The row corresponding to the fastest method in each group is highlighted in bold.

method	error	time	n_{iter}	n_{matvec}	m_{act}	speedup	method	error	time	n_{iter}	n_{matvec}	m_{act}	speedup
Arn	5.4e-12	12.5	5	750	150.0	0.0%	HmArn	1.3e-11	13.1	5	750	150.0	0.0%
sArn	**1.4e-11**	**8.9**	**5**	**750**	**150.0**	**28.6%**	**sHmArn**	**1.5e-11**	**9.2**	**5**	**750**	**150.0**	**30.0%**
adaArn-2	2.3e-7	39.5	325	4864	15.0	−216.1%	adaHmArn-2	3.6e-7	44.4	299	4784	16.0	−238.1%
adaArn-1	3.5e-7	36.0	365	4743	13.0	−188.1%	adaHmArn-1	3.2e-7	39.4	366	4758	13.0	−200.4%
adaArn-0	4.0e-7	52.8	638	5740	9.0	−322.4%	adaHmArn-0	7.6e-7	39.3	480	4320	9.0	−199.3%

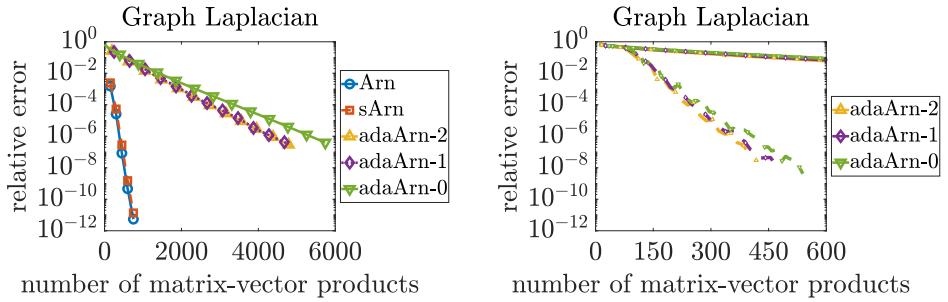


Fig. 3: Convergence for the fractional graph Laplacian example. Left: relative error versus the number of matrix-vector products without thick restarting. Right: adaptive Arnoldi methods without thick restarting (solid curves) and with thick restarting (dashed curves), shown over a smaller range of matrix-vector products. Fixed-dimension methods use $m = 150$ and $s = 4m = 600$; adaptive methods use $m_{\max} = 150$ and $\eta = 4$. The target retention dimension for thick restarting is 5, increased when necessary to preserve a conjugate pair.

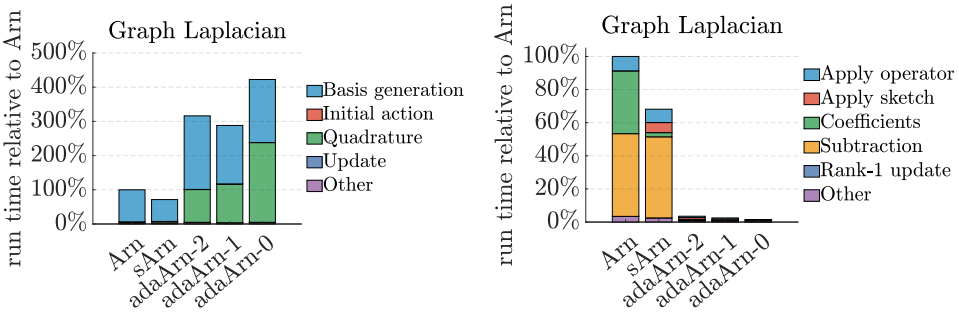


Fig. 4: Runtime breakdown for the fractional graph Laplacian example without thick restarting. The left panel shows the total runtime, normalized by that of Arn. The right panel shows the average basis-generation time per restart cycle, normalized by the corresponding average for Arn. Fixed-dimension methods use $m = 150$ and $s = 4m = 600$; adaptive methods use $m_{\max} = 150$ and $\eta = 4$.

Table 4: Adaptive Arnoldi methods with thick restarting for the graph Laplacian inverse square root example ($m_{\max} = 150$, $\eta = 4$). Columns report the relative error, runtime in seconds, number of restart cycles n_{iter}, number of matrix-vector products n_{matvec}, and mean actual Krylov subspace dimension per cycle m_{act}, including retained directions. Speedup is the percentage runtime reduction relative to Arn without thick restarting ($m = 150$), $(t_{\text{Arn}} - t)/t_{\text{Arn}} \times 100\%$.

method	error	time	n_{iter}	n_{matvec}	m_{act}	speedup
adaArn-2 (with thick restarting)	3.0e-8	3.3	28	418	19.8	73.4%
adaArn-1 (with thick restarting)	1.2e-8	3.4	37	481	17.9	72.9%
adaArn-0 (with thick restarting)	4.2e-9	3.2	60	539	13.9	74.6%

Without thick restarting, sArn and sHmArn are the fastest methods in their respective groups, with runtimes of 8.9 and 9.2 seconds, representing reductions of 28.6% and 30.0% relative to Arn and HmArn (Table 3). Basis generation still accounts for almost all the runtime of the fixed-dimension methods (Figure 4). Here, applying $\boldsymbol{A}$ is inexpensive, so reducing orthogonalization costs through sketching provides substantial savings. Only the Arnoldi variants are plotted, since the harmonic variants exhibit similar convergence and runtime breakdowns without thick restarting.

Without thick restarting, the adaptive methods require 299–638 restart cycles. Quadrature uses 128–256 nodes in most cycles after the initial stages and consequently accounts for a substantial fraction of the total runtime. The accumulated costs of basis generation and quadrature outweigh the savings per cycle, making these methods slower than the fixed-dimension methods.

Thick restarting substantially improves adaptive convergence, as shown in Figure 3 and Table 4. For the adaptive Arnoldi methods, it reduces the number of cycles from 325–638 to 28–60, the number of matrix-vector products from 4743–5740 to 418–539, and the runtime from 36.0–52.8 to 3.2–3.4 seconds. These methods are approximately 3.7–3.9 times as fast as Arn.

6.3. Convection-diffusion example. In this example, $\boldsymbol{A}$ is obtained by discretizing a two-dimensional convection-diffusion operator on the unit square with a constant convection field in the direction $[1, -1]$ and diffusion coefficient $\nu = 1$. More precisely,

$$\boldsymbol{A} = \frac{\nu}{h^2}\big(\boldsymbol{L} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{L}\big) + \frac{1}{h}\big(\boldsymbol{C} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{C}^\top\big) \in \mathbb{R}^{n^2 \times n^2},$$

where $h = 1/(n+1)$, $\boldsymbol{L} \in \mathbb{R}^{n\times n}$ is the tridiagonal matrix with diagonal entries 2 and sub- and superdiagonal entries -1, and $\boldsymbol{C} \in \mathbb{R}^{n\times n}$ is the matrix with diagonal entries 1 and subdiagonal entries -1. We aim to compute $f_2(\boldsymbol{A})\boldsymbol{b}$. In our experiments, n is set to be 512 so that $N = n^2 = 512^2 = 262144$, and $\boldsymbol{b} \in \mathbb{C}^N$ is a complex vector whose real and imaginary parts are drawn from the standard normal distribution. For the fixed-dimension methods, we set $m \in \{150, 200\}$ while for the adaptive methods, we set $m_{\max} = 200$.

Table 5: Relative error, runtime in seconds, number of restart cycles n_{iter}, number of matrix-vector products n_{matvec}, and average actual Krylov subspace dimension m_{act} per restart cycle for the convection-diffusion logarithm example. The column "speedup" reports the percentage runtime reduction $(t_{\text{base}} - t_{\text{curr}})/t_{\text{base}} \times 100\%$, where the baseline is Arn (left group) or HmArn (right group), using the fixed-dimension baseline at $m = m_{\max}$ for the adaptive section. For fixed-dimension methods, m is given in the section heading and $s = 4m$. Adaptive methods use $m_{\max} = 200$ and $\eta = 4$, and appear in a separate section.

method	error	time	n_{iter}	n_{matvec}	m_{act}	speedup	method	error	time	n_{iter}	n_{matvec}	m_{act}	speedup
$m = 150$, fixed													
Arn	1.9e-9	113.4	15	2250	150.0	0.0%	HmArn	5.1e-10	127.5	16	2400	150.0	0.0%
sArn	1.6e-9	86.2	19	2850	150.0	23.9%	sHmArn	1.8e-8	77.8	16	2400	150.0	39.0%
$m = 200$, fixed													
Arn	1.3e-10	144.3	11	2200	200.0	0.0%	HmArn	5.9e-11	177.3	13	2600	200.0	0.0%
sArn	5.0e-10	101.7	14	2800	200.0	29.5%	sHmArn	1.7e-8	97.4	13	2600	200.0	45.1%
$m_{\max} = 200$, adaptive													
adaArn-2	1.7e-10	84.1	11	2172	197.5	41.8%	adaHmArn-2	2.7e-10	98.6	12	2389	199.1	44.4%
adaArn-1	3.6e-7	184.4	594	10102	17.0	−27.8%	adaHmArn-1	6.1e-7	232.5	414	7855	19.0	−31.1%
adaArn-0	1.7e-6	320.7	2625	21000	8.0	−122.2%	adaHmArn-0	3.2e-6	352.7	2275	18200	8.0	−98.9%

Table 6: Adaptive Arnoldi methods with thick restarting for the convection-diffusion logarithm example ($m_{\max} = 200$, $\eta = 4$). Columns report the relative error, runtime in seconds, number of restart cycles n_{iter}, number of matrix-vector products n_{matvec}, and mean actual Krylov subspace dimension per cycle m_{act}, including retained directions. Speedup is the percentage runtime reduction relative to Arn without thick restarting ($m = 200$), $(t_{\text{Arn}} - t)/t_{\text{Arn}} \times 100\%$.

method	error	time	n_{iter}	n_{matvec}	m_{act}	speedup
adaArn-2 (with thick restarting)	4.7e-11	77.3	9	1771	201.2	46.5%
adaArn-1 (with thick restarting)	1.6e-7	80.5	203	3459	22.0	44.2%
adaArn-0 (with thick restarting)	7.7e-7	142.4	819	6552	13.0	1.3%

Table 5 shows that sketching substantially reduces the runtime. For $m = 150$ and

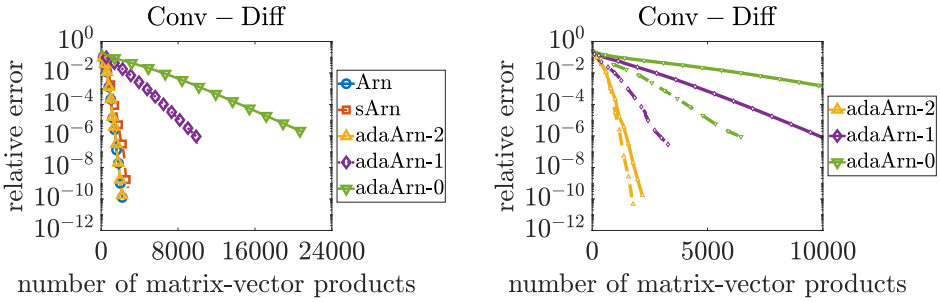


Fig. 5: Convergence for the convection-diffusion example. Left: relative error versus the number of matrix-vector products without thick restarting. Right: adaptive Arnoldi methods without thick restarting (solid curves) and with thick restarting (dashed curves), shown over a smaller range of matrix-vector products. Fixed-dimension methods use $m = 200$ and $s = 4m = 800$; adaptive methods use $m_{\max} = 200$ and $\eta = 4$. The target retention dimension for thick restarting is 5, increased when necessary to preserve a conjugate pair.

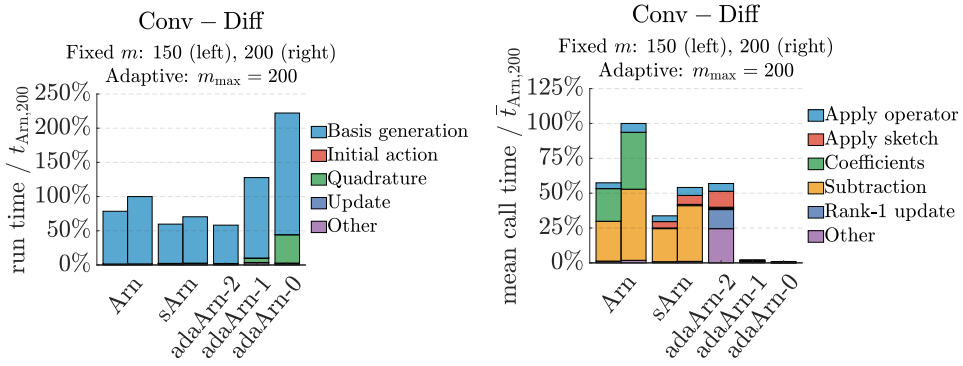


Fig. 6: Runtime breakdown for the convection-diffusion example without thick restarting. Left: total runtime, normalized by that of Arn with $m = 200$. Right: average basis-generation time per restart cycle, normalized by the corresponding average for Arn with $m = 200$. For each fixed-dimension method, the left and right bars correspond to $m = 150$ and $m = 200$, respectively, with $s = 4m$. Adaptive methods use $m_{\max} = 200$ and $\eta = 4$.

$m = 200$, sArn reduces the total runtime relative to Arn by 23.9% and 29.5%, respectively, while sHmArn achieves reductions of 39.0% and 45.1% relative to HmArn. Here, applying $\boldsymbol{A}$ is inexpensive, so orthogonalization accounts for a large fraction of the basis-generation cost. Sketching reduces this cost substantially, as illustrated in Figure 6.

The average basis-generation time per cycle is reduced by 41.2% for $m = 150$ and 45.9% for $m = 200$. These savings do not translate directly into comparable reductions in total runtime, because sArn requires more restart cycles than Arn: 19

rather than 15 for $m = 150$, and 14 rather than 11 for $m = 200$. For example, at $m = 200$, the ratio of the accumulated basis-generation times is approximately

$$\frac{T_{\text{basis,sArn}}}{T_{\text{basis,Arn}}} = \frac{6.99}{12.92}\,\frac{14}{11} \approx 0.689.$$

Thus, the additional cycles offset part of the savings per cycle, leaving a reduction of approximately 31.1% in the accumulated basis-generation time. Including the remaining costs gives the observed total runtime reduction of 29.5%.

Increasing m from 150 to 200 reduces the number of restart cycles for all four fixed-dimension methods, but increases their total runtimes. For Arn and sArn, the number of matrix-vector products decreases only slightly, from 2250 to 2200 and from 2850 to 2800, respectively. For HmArn and sHmArn, it increases from 2400 to 2600. The larger subspaces therefore make each cycle more expensive without providing sufficient improvement in convergence to compensate. Although the relative savings from sketching are greater at $m = 200$, all four fixed-dimension methods are faster at $m = 150$.

Among the methods with $m = m_{\max} = 200$ without thick restarting, adaArn-2 is the fastest, requiring 84.1 seconds and 2172 matrix-vector products. It also uses the fewest matrix-vector products among all configurations in Table 5, although sHmArn with $m = 150$ has a shorter runtime of 77.8 seconds. In contrast, adaArn-1 and adaArn-0 generate much smaller subspaces but require many more restart cycles, making them slower than Arn.

Thick restarting substantially improves the convergence of adaArn-1 and adaArn-0; see Figure 5 and Table 6. Their matrix-vector counts decrease from 10102 to 3459 and from 21000 to 6552, while their runtimes decrease from 184.4 to 80.5 seconds and from 320.7 to 142.4 seconds, respectively. Nevertheless, adaArn-0 remains only 1.3% faster than Arn with $m = 200$. The benefit is more modest for adaArn-2, whose runtime decreases from 84.1 to 77.3 seconds and whose matrix-vector count decreases from 2172 to 1771. With thick restarting, adaArn-2 is the fastest among all configurations reported here and requires the fewest matrix-vector products.

7. Conclusions. We have developed a sketch-and-restart framework for computing $f(\boldsymbol{A})\boldsymbol{b}$ with large sparse non-Hermitian matrices. The framework combines quadrature-based restarting with Arnoldi-like decompositions generated by sketched or truncated Krylov processes. The key observation is that several Krylov and sketched Krylov approximations share the common form $\boldsymbol{B}_m f(\boldsymbol{H}_m)\boldsymbol{e}_1\beta$, which leads to a unified error representation and restarting mechanism. Within this framework, we combined the existing s-Arnoldi process with quadrature-based restarting and introduced a new sHm-Arnoldi process together with its restarted variant. We further developed adaptive restarted Arnoldi and HmArnoldi methods based on the t-Arnoldi process and a rank-1 update. For Stieltjes functions of positive real matrices, we derived conditional convergence bounds for the restarted sHmArnoldi method under suitable embedding and sketched harmonic Ritz-value assumptions, and provided a sufficient condition ensuring that these Ritz values lie in the open right half-plane. More broadly, the proposed framework separates the restarting mechanism from the basis generation procedure, making it possible to incorporate other randomized Krylov processes within the same framework.

Numerical experiments show that sketching reduces orthogonalization costs and total runtime in the reported examples. The adaptive methods select the restart length automatically and can substantially reduce basis storage, although the smaller

subspaces may require more restart cycles and increase the total runtime. For the adaptive Arnoldi variants, thick restarting substantially improves convergence in the graph Laplacian and convection–diffusion examples, while retaining small subspaces under aggressive truncation. The use of BLAS-2 operations in the final rank-1 update also motivates further investigation of high-performance implementations.

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